Teaching Reproducible Research Using R

Christian Martinez

Table of contents

About This Book 10
- What This Book Represents . . . 10
- How This Book Is Organized . . . 11
- Reproducibility as a Professional Skill . . . 11
- From Assignment to Authorship . . . 12
- License . . . 12
  - How to Cite . . . 13
- A Model of Publication in Practice . . . 13
- A Note to Readers . . . 14

Course Blueprint 15
- Course Philosophy . . . 15
- Learning Objectives . . . 15
- Course Structure . . . 16
  - Tuesdays — Concept + Guided Implementation . . . 16
  - Thursdays — Applied Workshop . . . 16
- Assessment Design . . . 16
  - Weekly Assignments (40%) . . . 17
  - Midterm Project (30%) . . . 17
  - Final Project (30%) . . . 17
- Conceptual Arc of the Semester . . . 18
  - Foundations . . . 18
  - Visualization & Exploration . . . 18
  - Statistical Inference . . . 18
  - Reproducible Communication . . . 19
- Open Science Integration . . . 19
- From Coursework to Authorship . . . 20
- Institutional Policies & Resources . . . 20
- Closing Note . . . 21

1 Base R Foundations 22
- 1.1 Overview . . . 22
- 1.2 Learning Objectives . . . 22
- 1.3 Textbook Connection . . . 22
- 1.4 Submission Instructions . . . 23

1.5 Assignment Tasks . . . 23
1.5.1 1. R as a Calculator . . . 23
1.5.2 2. Vectors . . . 23
1.5.3 3. Data Frames . . . 24
1.5.4 4. Built-in Datasets . . . 24
1.5.5 Reflection . . . 24
1.6 Reproducibility Practice . . . 25

2 Exploring Pixar with the Tidyverse 26
2.1 Overview . . . 26
2.2 Learning Objectives . . . 26
2.3 Textbook Connection . . . 27
2.4 Submission Instructions . . . 27
2.5 Assignment Tasks . . . 27
2.5.1 Exploring Pixar Genres . . . 27
2.5.2 Which Pixar Films Made the Most Money? . . . 28
2.5.3 Turning `cinema_score` Grades into Numbers . . . 29
2.5.4 Pixar Through the Decades . . . 30
2.5.5 Pixar Data Challenge . . . 31
2.6 Reflection . . . 31
2.7 Reproducibility Practice . . . 32

3 NYPD Shooting Incidents — Cleaning, Insights, and Visualization 33
3.1 Overview . . . 33
3.2 Learning Objectives . . . 33
3.3 Textbook Connection . . . 34
3.4 Submission Instructions . . . 34
3.5 Assignment Tasks . . . 34
3.5.1 Data Source . . . 34
3.5.2 Cleaning and Feature Engineering . . . 34
3.5.3 Insights . . . 36
3.5.4 Visualizations (`ggplot2`) . . . 36
3.5.5 Written Summary . . . 37
3.6 Reproducibility Practice . . . 37

4 Merging Data and Comparing Means with t-tests 38
4.1 Overview . . . 38
4.2 Learning Objectives . . . 38
4.3 Textbook Connection . . . 39
4.4 Submission Instructions . . . 39
4.5 Assignment Tasks . . . 39
4.5.1 1. Data Wrangling (Import + Merge) . . . 39
4.5.2 2. Handling Missing Data (Imputation) . . . 40

4.5.3 3. Dependent t-test (Paired) . . . 41
4.5.4 4. Independent t-tests (Sex + Department) . . . 41
4.5.5 Written Summary . . . 42
4.6 Reproducibility Practice . . . 42

5 R Script → Quarto Report 43
5.1 Overview . . . 43
5.2 Learning Objectives . . . 43
5.3 Textbook Connection . . . 43
5.4 Submission Instructions . . . 44
5.5 Assignment Tasks . . . 44
5.5.1 1. YAML header . . . 44
5.5.2 2. Setup chunk . . . 44
5.5.3 3. Minimum required code chunks (4) . . . 44
5.5.4 4. Narrative text . . . 45
5.5.5 5. Inline R expression . . . 45
5.5.6 6. Hyperlink to the dataset . . . 45
5.5.7 Reflection . . . 46
5.6 Publishing Instructions . . . 46
5.7 Reproducibility Practice (Quarto Reporting) . . . 46

6 ANOVA and Payment Patterns in NYC Camera Violations 47
6.1 Overview . . . 47
6.2 Learning Objectives . . . 47
6.3 Textbook Connection . . . 48
6.4 Submission Instructions . . . 48
6.5 Assignment Tasks . . . 48
6.5.1 1. Data Ingest . . . 49
6.5.2 2. Data Cleaning and Feature Engineering . . . 49
6.5.3 3. Analysis 1: Day of Week and Fine Amount . . . 50
6.5.4 4. Analysis 2: Time of Day and Fine Amount . . . 51
6.5.5 5. Analysis 3: Violation Type and Fine Amount . . . 51
6.5.6 6. Analysis 4: Violation Type and Payment Amount . . . 52
6.5.7 Final Reflection . . . 52
6.6 Publishing Instructions . . . 53
6.7 Reproducibility Practice . . . 53
6.8 Citing Your R Packages . . . 54
6.8.1 Your Task . . . 54

7 Midterm: Exercise & Sleep 55
7.1 Overview . . . 55
7.2 Learning Objectives . . . 55
7.3 Textbook Connection . . . 56

7.4 Submission Instructions . . . 56
7.5 Quarto & Reproducibility Requirements . . . 56
7.5.1 1. YAML & Document Organization . . . 57
7.5.2 2. Code Chunk Labels . . . 57
7.5.3 3. Figures, Tables & Cross-References . . . 57
7.5.4 4. Inline R Code . . . 58
7.5.5 5. Citing Your R Packages . . . 58
7.6 Midterm Tasks . . . 59
7.6.1 1. Setup & Data Import . . . 59
7.6.2 2. Merge & Base Cleaning . . . 59
7.6.3 3. Create Derived Variables . . . 60
7.6.4 4. Descriptive Statistics . . . 61
7.6.5 5. Visualizations (3 Plots) . . . 61
7.6.6 6. t-tests (Two) . . . 62
7.6.7 7. ANOVAs (Two) + Post-hoc Tests . . . 63
7.6.8 8. Synthesis & Recommendation . . . 63
7.6.9 9. Reflection . . . 64
7.7 Publishing Instructions . . . 65
7.8 Reproducibility Practice . . . 65

8 CUNY Analytics — Exploring Team Performance Through Reproducible Analysis 66
8.1 Overview . . . 66
8.2 Learning Objectives . . . 66
8.3 Textbook Connection . . . 67
8.4 Submission and Formatting Requirements . . . 67
8.5 Assignment Tasks . . . 68
8.5.1 Retrieve and Explore the Data . . . 68
8.5.2 Prepare Brooklyn College Men's Basketball Data . . . 68
8.5.3 Visual Exploration . . . 69
8.5.4 Correlation Analysis . . . 70
8.5.5 Your Independent Research Question . . . 71
8.6 Executive Memo . . . 71
8.7 Reflection and Looking Ahead . . . 72
8.7.1 Reproducibility Practice . . . 72
8.8 Final Checklist . . . 73

9 CUNY Analytics — From Correlation to Regression 74
9.1 Overview . . . 74
9.2 Learning Objectives . . . 74
9.3 Textbook Connection . . . 75
9.4 Submission and Formatting Requirements . . . 75
9.5 Assignment Tasks . . . 76
9.5.1 Continuing Your Investigation . . . 76

9.5.2 Simple Linear Regression . . . 76
9.5.3 Predictions, Residuals, and Model Diagnostics . . . 77
9.5.4 Multiple Regression and Model Comparison . . . 78
9.5.5 Did Regression Change Your Conclusions? . . . 79
9.6 Executive Memo . . . 79
9.7 Reflection . . . 80
9.8 Publishing and Reproducibility . . . 80
9.9 Final Checklist . . . 81

10 Streaming Analytics — Understanding Platform Popularity Across Age Groups 82
10.1 Overview . . . 82
10.2 Learning Objectives . . . 83
10.3 Textbook Connection . . . 83
10.4 Submission Instructions . . . 83
10.5 Assignment Tasks . . . 84
10.5.1 1. Data Preparation . . . 84
10.5.2 2. Visualization . . . 84
10.5.3 3. Chi-Square Test of Independence . . . 85
10.5.4 4. Observed, Expected, and Residual Values . . . 85
10.5.5 5. Contributions to the Chi-Square Statistic . . . 86
10.5.6 6. Effect Size (Cramer's V) . . . 86
10.5.7 7. Final Interpretation . . . 87
10.6 Deliverables . . . 87
10.7 Guidelines for Interpretation . . . 87
10.8 Reflection . . . 88
10.9 Publishing Instructions . . . 88
10.10Reproducibility Practice . . . 88

11 Wage Analytics — Predicting High vs Low Earners Using Logistic Regression 89
11.1 Overview . . . 89
11.2 Learning Objectives . . . 89
11.3 Textbook Connection . . . 90
11.4 Submission Instructions . . . 90
11.5 Assignment Tasks . . . 90
11.5.1 1. Create the WageCategory Variable . . . 90
11.5.2 2. Data Cleaning . . . 91
11.5.3 3. Classical Statistical Tests . . . 91
11.5.4 4. Logistic Regression Model . . . 93
11.5.5 5. Model Evaluation on Test Data . . . 94
11.5.6 6. Final Interpretation . . . 94
11.6 Guidelines for Interpretation . . . 95
11.7 Reflection . . . 95
11.8 Publishing Instructions . . . 95

11.9 Reproducibility Practice . . . 96

12 Final Project: NYC Open Data 2026 Proposal **97**
12.1 Overview . . . 97
12.2 Learning Objectives . . . 97
12.3 Textbook Connection . . . 98
12.4 Project Requirements . . . 98
12.4.1 1. Dataset Selection . . . 98
12.4.2 2. Reproducible Analysis (Quarto) . . . 99
12.4.3 3. Proposal-Style Structure . . . 99
12.5 Length & Formatting Requirements . . . 101
12.6 Class Presentation . . . 102
12.7 Class Publication (Optional but Encouraged) . . . 102
12.8 Grading (100 Points Total) . . . 103
12.9 Reproducibility Practice . . . 103
12.10 Resources . . . 104
12.11 Submission . . . 104
12.12 Takeaway . . . 104

13 Final Assignment: Building Your Reproducible Quarto Book **105**
13.1 Overview . . . 105
13.2 Learning Objectives . . . 106
13.3 Textbook Connection . . . 106
13.4 Assignment Requirements . . . 106
13.4.1 1. Create a Quarto Book Project . . . 106
13.4.2 2. Required Chapters (In This Order) . . . 107
13.4.3 3. Title Page . . . 107
13.4.4 4. Cover Image . . . 108
13.4.5 5. Chapter Introductions . . . 108
13.4.6 6. Figure and Table Captions . . . 108
13.4.7 7. Final Reflection Chapter . . . 109
13.5 Book Quality Standards . . . 109
13.6 Publishing Requirement . . . 110
13.7 Grading (100 Points Total) . . . 110
13.8 Reproducibility Practice . . . 111
13.9 Takeaway . . . 111

Appendices **112**

A Undergraduate Midterm: Exercise & Sleep **112**
A.1 Overview . . . 112
A.2 Learning Objectives . . . 112

A.3 Textbook Connection . . . 113
A.4 Submission Instructions . . . 113
A.5 Quarto & Reproducibility Requirements . . . 113
A.5.1 1. YAML & Document Organization . . . 114
A.5.2 2. Code Chunk Labels . . . 114
A.5.3 3. Figures, Tables & Cross-References . . . 114
A.5.4 4. Inline R Code . . . 115
A.5.5 5. Citing Your R Packages . . . 115
A.6 Midterm Tasks . . . 116
A.6.1 1. Setup & Data Import . . . 116
A.6.2 2. Merge & Base Cleaning . . . 116
A.6.3 3. Create a New Variable . . . 117
A.6.4 4. Descriptive Statistics . . . 118
A.6.5 5. Visualizations (2 Plots) . . . 118
A.6.6 6. t-tests . . . 119
A.6.7 7. Synthesis & Recommendation . . . 120
A.6.8 8. Reflection . . . 120
A.7 Publishing Instructions . . . 121
A.8 Reproducibility Practice . . . 121

B Assignment 6: ANOVA and Payment Patterns in NYC Camera Violations 122
B.1 Overview . . . 122
B.2 Why Are We Learning ANOVA in Psychology? . . . 122
B.3 Learning Objectives . . . 123
B.4 Textbook Connection . . . 124
B.5 Submission Instructions . . . 124
B.6 Part 1: Load and Explore the Data . . . 125
B.6.1 Your Tasks . . . 125
B.6.2 Questions . . . 125
B.7 Part 2: Prepare Our Data . . . 125
B.8 2A. Convert Numeric Variables . . . 126
B.9 2B. Create Day of Week . . . 126
B.10 Need a Hint? . . . 126
B.10.1 Quick Check . . . 127
B.11 Part 3: ANOVA Investigation 1 — Fine Amount by Day of Week . . . 127
B.12 Step 1: Identify Your Variables . . . 127
B.13 Step 2: Descriptive Statistics . . . 127
B.14 Step 3: Conduct a One-Way ANOVA . . . 128
B.15 ANOVA Reminder . . . 128
B.15.1 Interpret Your Results . . . 128
B.16 Statistical Significance vs. Practical Importance . . . 128
B.17 Step 4: Explore Post-Hoc Comparisons . . . 129
B.18 Step 5: Interpret Your Findings . . . 129

B.19 Part 4: ANOVA Investigation 2 — Payment Amount by Violation Type 129
B.20 Your Tasks 130
B.20.1 Interpret Your Findings 130
B.21 Think Like a Researcher! 131
B.22 Part 5: Final Reflection 131
B.23 Part 6: Citing Your R Packages 132
B.23.1 Your Tasks 132
B.23.2 Need a Reminder? 132
B.23.3 Example: Generating a Bibliography 133
B.24 Final Checklist 134

C References **135**

D References **136**

About This Book

This book represents the applied companion to Reproducible Research Using R.

Over the course of the semester, students transformed structured assignments into fully reproducible analyses using real datasets, transparent workflows, and public-facing communication. What began as individual homework submissions evolved into polished research chapters assembled into this cohesive volume.

This book treats reproducibility not as an enhancement, but as a default professional practice.

Each chapter reflects a progression of skills: data import, cleaning, visualization, statistical testing, modeling, interpretation, and communication. More importantly, each chapter reflects a shift in mindset — from "running code" to designing complete analytical workflows.

This is not just a collection of assignments.
It is a curated portfolio of applied reproducible research.

What This Book Represents

This volume showcases:

- Applied statistical reasoning
- Transparent data cleaning and transformation
- Reproducible modeling and interpretation
- Clear, professional communication of results
- Public-facing research using real-world datasets

Every analysis in this book can be rerun from start to finish. All figures, tables, and results are generated directly from executable code. Nothing has been manually edited or pasted from external software.

The goal is not perfection. The goal is transparency.

How This Book Is Organized

The chapters follow the structure of the course and mirror the conceptual arc of *Reproducible Research Using R*.

- Base R Foundations
- mtcars Wrangling and Feature Engineering
- NYPD Shooting Incidents — Cleaning, Insights, and Visualization
- Merging Data and Comparing Means with t-tests
- R Script → Quarto Report
- ANOVA and Payment Patterns in NYC Camera Violations
- Midterm: Exercise & Sleep – Experimental Analysis and Inference
- NBA Analytics – Correlation and Partial Correlation
- Florida Crime Analytics – Regression Modeling and Model Comparison
- Streaming Analytics – Categorical Analysis and Chi-Square Testing
- Wage Analytics – Logistic Regression and Predictive Modeling
- Final Project – Independent Civic Data Research
- Final Assignment – Building Your Reproducible Quarto Book

Each chapter applies a specific methodological framework to a real dataset.

Together, they form a complete analytical portfolio.

Reproducibility as a Professional Skill

Throughout this book, reproducibility is treated as a default practice rather than an optional enhancement.

Each chapter:

- Loads data programmatically

- Cleans variables transparently
- Documents transformation decisions
- Avoids hard-coded statistical results
- Produces outputs directly from executable code

This workflow ensures that analyses are inspectable, repeatable, and defensible — qualities expected in graduate research, industry analytics, and public-facing data work.

From Assignment to Authorship

By assembling these projects into a Quarto book, students move beyond submission-based coursework and into publication-based thinking.

This book serves as:

- A professional portfolio artifact
- A demonstration of applied statistical competency
- Evidence of reproducible workflow mastery
- A foundation for conference submissions and civic engagement

Completing this book is not simply a course requirement.

It is a transition point — from student to analytical author.

License

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How to Cite

If you use this textbook in your teaching, research, or projects, please cite it as:

Martinez, C. (2026). *Teaching Reproducible Research Using R* (Version 1.0.1). Zenodo. https://doi.org/10.5281/zenodo.19139302

Version History

v1.0.1 (2026)

- Initial public release
- Full textbook curriculum covering data analysis, visualization, and modeling in R
- Used in Fall 2025 coursework at Brooklyn College (CUNY)

A Model of Publication in Practice

A previous cohort of students transformed their final projects into a collective publication: NYC Open Data Student Gallery - Brooklyn College

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A Note to Readers

If you are a future employer, graduate program, research lab, or collaborator reviewing this book, you are not just reading assignments — you are reviewing a semester-long demonstration of analytical growth.

If you are a student currently enrolled in the course, this book represents what is possible when structured workflow and intentional practice are sustained over time.

Reproducibility is not a feature of advanced researchers.
It is a habit formed early.

Course Blueprint

Instructor: Christian A. Martinez
Institution: Brooklyn College, CUNY

Course Philosophy

This course is structured around a simple premise: reproducibility is not an advanced skill — it is a foundational habit.

Rather than treating R as a programming language to learn in isolation, this course approaches R as a complete research environment. Students learn how to move from raw data to interpretable results using transparent, inspectable, and *reproducible* workflows.

By the end of the semester, students do not just complete assignments — they produce a cohesive analytical portfolio.

Learning Objectives

By the end of this course, students are able to:

- Write and execute R scripts for applied data analysis
- Clean and wrangle datasets using reproducible workflows
- Create publication-quality visualizations using `ggplot2`
- Conduct and interpret common statistical analyses (t-tests, ANOVA, regression, logistic regression, Chi-Square)

- Reproduce analyses from published psychology papers with open datasets
- Communicate results in structured Quarto documents
- Apply open science practices (e.g., OSF, GitHub, transparent documentation)
- Assemble individual analyses into a polished Quarto book

Course Structure

The course follows a consistent weekly rhythm designed to reinforce applied learning.

Tuesdays — Concept + Guided Implementation

- Short conceptual lecture (20–25 minutes)
- Guided mini-lab exercises
- Emphasis on understanding the logic behind each method

Thursdays — Applied Workshop

- Live coding demonstration
- Hands-on work with real datasets
- Reproduction of analyses from published psychology papers
- Debugging as a collaborative process
- Screen-sharing and peer problem solving encouraged

This structure ensures that students immediately apply conceptual material in authentic research contexts.

Assessment Design

Evaluation in this course is cumulative and portfolio-driven.

Weekly Assignments (40%)

Each week introduces a new methodological framework. Assignments require students to:

- Clean and analyze real datasets
- Interpret statistical output
- Produce reproducible visualizations
- Write structured narrative explanations

Assignments progressively increase in complexity.

Midterm Project (30%)

The midterm integrates:

- Data import and merging
- Variable engineering
- t-tests and ANOVA
- Post-hoc comparisons
- Evidence-based recommendation

Students demonstrate their ability to conduct a complete inferential workflow independently.

Final Project (30%)

Students design and conduct an original research project using NYC Open Data or other civic datasets.

The final submission includes:

- Fully reproducible analysis
- Clear research framing
- Professional visualizations
- Public-facing interpretation
- In-class presentation

Conceptual Arc of the Semester

The course progresses through four major phases:

Foundations

- R basics
- Data structures
- Functions and packages
- Tidyverse workflows

Visualization & Exploration

- `ggplot2`
- Feature engineering
- Exploratory data analysis
- Data cleaning decisions

Statistical Inference

- t-tests
- ANOVA
- Correlation
- Linear regression
- Chi-Square tests
- Logistic regression

Reproducible Communication

- Quarto documents
- Structured reporting
- Transparent code practices
- Building a Quarto book

Each stage builds directly on the previous one.

Open Science Integration

Reproducibility is treated as a default expectation.

Students are required to:

- Load data programmatically
- Avoid manual spreadsheet edits
- Document all cleaning decisions
- Avoid hard-coding statistical results
- Use structured document formats
- Present analyses transparently

Open science practices such as preregistration, OSF usage, and GitHub version control are introduced and encouraged throughout the semester.

From Coursework to Authorship

The final assignment requires students to assemble all homeworks into a cohesive Quarto book, moving away from "one-and-done" assignments.

This transforms:

- Discrete homework submissions
  into
- A structured analytical portfolio

The result is a professional artifact suitable for:

- Graduate school applications
- Research lab submissions
- Industry analytics portfolios
- Conference presentations
- Public-facing civic research

The structure of the course ensures that students leave not only with technical fluency in R, but with a coherent body of *reproducible* work.

Institutional Policies & Resources

The course follows all Brooklyn College and CUNY policies regarding academic integrity, accessibility services, and student support resources.

Students are expected to adhere to the CUNY Academic Integrity Policy and to utilize campus support services when needed.

For official policy details, students should refer to the Brooklyn College website and Undergraduate Bulletin.

Closing Note

Reproducible research is not about perfection.

*It is about **clarity**.*
*It is about **transparency**.*
*It is about building **analytical habits** that scale beyond a single course.*

This blueprint reflects a semester-long progression from foundational skills to independent analytical authorship.

1 Base R Foundations

1.1 Overview

This assignment introduces core concepts in base R, including arithmetic operations, object assignment, vectors, data frames, and built-in datasets. The goal is to build comfort with fundamental R syntax before moving into more advanced workflows.

You will complete all tasks in a single `.R` script.

1.2 Learning Objectives

By completing this assignment, you will be able to:

- Perform arithmetic operations in R
- Assign results to objects using `<-`
- Create and manipulate vectors
- Construct and subset data frames
- Explore built-in datasets
- Produce a simple base R visualization
- Organize a script using clear comments

1.3 Textbook Connection

This assignment builds directly from **Chapter 1: Introduction to R** in Reproducible Research Using R.

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

1.4 Submission Instructions

Submit **one `.R` script file**.

Your script must:

- Be clearly organized into labeled sections
  (e.g., `# Part 1: R as a Calculator`)
- Include all required calculations and outputs
- Include your reflection as comments at the end
- Run from top to bottom without errors

1.5 Assignment Tasks

1.5.1 1. R as a Calculator

1. Calculate `12 * 2`.
2. Calculate the square root of `144`.
3. Calculate `100 / 7` and round to **2 decimals**.
4. Save the result of `15 * 4` as `my_product`.
5. Save `100 / 7`, rounded to **5 decimals**, as `my_second_product`.

1.5.2 2. Vectors

1. Create a numeric vector of five random numbers named `randy`.
2. Compute the mean, median, minimum, and maximum of `randy`.
3. Create another numeric vector of five random numbers named `randy2`.
4. Create `randy_combined` by adding `randy` and `randy2`.
5. Create a character vector of five favorite movies.
6. Use indexing to print the fourth movie.
7. Create a factor with 11 entries of favorite foods (repeat some).
8. Use `table()` to count frequencies.
9. Sort the table in descending order and save as `organized_table`.

1.5.3 3. Data Frames

1. Create a data frame called `class_df` with columns:
    - name
    - age
    - major
2. Add at least four fictional classmates.
3. Create `age_in_2009`.
4. Create `favorite_professor`.
5. Use indexing to:
    - Select only name and major.
    - Filter rows for Psych majors.

1.5.4 4. Built-in Datasets

Using `mtcars`:

1. Display the first 11 rows.
2. List all column names.
3. Run `str(mtcars)` and describe column types in a comment.
4. Run `summary(mtcars)`.
5. Create a base R scatterplot titled:

   "My First Scatterplot"

1.5.5 Reflection

At the end of your script, answer in 2–3 sentences (as comments):

What was the most surprising or confusing thing you learned about R so far?

1.6 Reproducibility Practice

For this first assignment, focus on script organization and documentation.

At the top of your script, include a structured comment block containing:

- Assignment name
- Your name
- Date
- A brief description of what the script accomplishes

Your script should:

- Be divided into clearly labeled sections
- Run from top to bottom without manual intervention
- Avoid unnecessary objects

This assignment establishes your baseline reproducibility habits for the semester.

2 Exploring Pixar with the Tidyverse

2.1 Overview

This assignment continues our introduction to the **Tidyverse** using data from the `pixarfilms` package.

The `pixarfilms` package contains several datasets describing Pixar movies, including their genres, box-office performance, audience responses, and release information.

You will use these datasets to answer a series of questions about Pixar's history and films.

2.2 Learning Objectives

By completing this assignment, you will be able to:

- Use `dplyr` verbs such as `select()`, `distinct()`, `mutate()`, `count()`, `group_by()`, `summarize()`, and `arrange()` to explore data
- Identify unique values within a dataset
- Create new variables from existing variables
- Use `case_when()` to recode categorical information
- Identify the highest and lowest values using `slice_max()` and `slice_min()`
- Calculate grouped summary statistics
- Apply Tidyverse skills to a new dataset independently
- Develop and answer your own data-driven question

2.3 Textbook Connection

This assignment builds directly from **Chapter 2: Introduction to tidyverse** in *Reproducible Research Using R*.

Students are encouraged to review the chapter and the code we worked through together in class before beginning this assignment.

2.4 Submission Instructions

Submit **one `.R` script file**.

Your script must:

- Run from top to bottom without errors
- Include clear section comments for each question
- Save newly created datasets when instructed
- Contain your written responses as comments where requested

Begin your script by loading the packages needed for this assignment:

```
library(tidyverse)
library(pixarfilms)
```

2.5 Assignment Tasks

2.5.1 Exploring Pixar Genres

The `genres` dataset contains the genres associated with each Pixar film. Notice that a movie can appear more than once because a single movie may belong to multiple genres.

2.5.1.1 1.1 What Genres Appear in Pixar Films?

First, let's find out which genres appear in the dataset.

Using the `genres` dataset:

1. Use `distinct()` to identify the unique genres represented in Pixar films.
2. Determine the **total number of distinct genres**.
3. Arrange the unique genres alphabetically.

Question: How many distinct genres appear in the dataset?

2.5.1.2 1.2 How Many Genres Does Each Film Have?

Next, investigate how many genres are associated with each Pixar film.

Using the `genres` dataset:

1. Count the number of genres associated with each film.
2. Arrange the results from the **greatest number of genres to the fewest**.
3. Save the result as `film_genres`.

Question: Which Pixar film or films are associated with the greatest number of genres?

2.5.2 Which Pixar Films Made the Most Money?

The `box_office` dataset contains information about the production budget and box-office performance of Pixar films.

For this assignment, we will estimate how much money a film made by subtracting its production budget from its worldwide box-office revenue.

Using `box_office`:

1. Use `select()` to keep only the following variables:
   - `film`
   - `budget`
   - `box_office_worldwide`
2. Use `mutate()` to create a new variable called `net`, calculated as:

   worldwide box office - budget

3. Save the resulting dataset as `pixar_money`.
4. Using `slice_max()`, identify the **three films with the highest net**.
5. Using `slice_min()`, identify the **three films with the lowest net**.

Questions:

- Which Pixar film had the highest estimated net?
- Which Pixar film had the lowest estimated net?

> **Note**
>
> For this assignment, `net` is a simplified measure of financial performance. A film's production budget and worldwide box-office revenue do not account for all costs or sources of revenue.

2.5.3 Turning `cinema_score` Grades into Numbers

The `public_response` dataset contains several measures of how Pixar films were received by critics and audiences.

One of these variables is `cinema_score`, which uses letter grades such as `"A+"`, `"A"`, `"A-"`, and so on.

Let's convert these letter grades into numeric scores.

Using the `public_response` dataset:

1. Use `mutate()` and `case_when()` to create a new variable called `cinema_numeric`.
2. Use the **grading scale from our course syllabus** to determine the numeric value associated with each letter grade.
3. Save the resulting dataset as `pixar_scores`.
4. Calculate the **average numeric** `cinema_score` across Pixar films.
5. Identify the **three films with the highest numeric** `cinema_score`.
6. Identify the **three films with the lowest numeric** `cinema_score`.

Questions:

- What was the average numeric `cinema_score` for Pixar films?
- Which films received the highest converted scores?
- Which films received the lowest converted scores?

2.5.4 Pixar Through the Decades

The `pixar_films` dataset contains basic information about Pixar's films, including their release dates and runtimes.

We want to investigate how Pixar's films have changed across decades.

2.5.4.1 4.1 Create a Decade Variable

Using `pixar_films`, use `mutate()` to create a new variable called `decade` based on `release_date`.

Your categories should identify films as belonging to the:

- 1990s
- 2000s
- 2010s
- 2020s

Save the resulting dataset as `pixar_decades`.

2.5.4.2 4.2 Summarize Pixar Films by Decade

Next, group the films by `decade` and calculate:

- Number of films
- Mean runtime
- Median runtime
- Minimum runtime
- Maximum runtime

Arrange the resulting table from the **most recent decade to the oldest**.

Save this summary as `decade_summary`.

Question: Based on your summary, do Pixar movies appear to have become longer, shorter, or stayed approximately the same length over time? Briefly explain your answer as a comment in your script.

2.5.5 Pixar Data Challenge

Now it's your turn!

Using **any dataset in the `pixarfilms` package**, come up with **one question about Pixar that you find interesting**.

Some possibilities include questions about:

- Box-office performance
- Genres
- Movie ratings
- Academy Awards
- Release dates
- Runtime

You are not limited to these topics.

Write your question as a comment in your script.

Then:

1. Use at least **three Tidyverse functions** to investigate your question.
2. Show the code used to answer it.
3. Write **1–2 sentences as comments** explaining what you found.

2.6 Reflection

At the end of your script, answer the following questions as comments:

- Which part of this assignment felt the hardest for you? Why?
- Which part felt the easiest for you? Why?
- Which Pixar film is your favorite? *(There is no wrong answer… probably.)*

2.7 Reproducibility Practice

This assignment focuses on creating a **clear and reproducible data-analysis workflow**.

In your script:

- Use section headers that match Questions 1–5.
- Give newly created objects clear and meaningful names.
- Do not overwrite the original datasets from the `pixarfilms` package.
- Use `View()`, `glimpse()`, `summary()`, `count()`, or other checkpoint functions when helpful to inspect your work.
- Make sure your completed script runs from **top to bottom without errors**.

Remember: the goal is not simply to get an answer. Your code should make it possible for someone else to understand **how you got there**.

3 NYPD Shooting Incidents — Cleaning, Insights, and Visualization

3.1 Overview

In this assignment, you will work with **NYPD Shooting Incident Data** using the `nycOpenData` package to practice data cleaning, feature engineering, exploratory insight generation, and data visualization with `ggplot2`.

Compared to previous assignments, this one is more open-ended: you will make a few analytic decisions (e.g., how to define time-of-day categories and which variables to explore for your second insight). This marks your transition from structured exercises to defensible analytic decision-making.

3.2 Learning Objectives

By completing this assignment, you will be able to:

- Retrieve real-world NYC civic data using `nycOpenData`
- Identify and handle missing data in a defensible way
- Engineer new features from dates/times and categorical columns
- Generate exploratory insights using grouping and summaries
- Communicate findings using well-designed `ggplot2` visualizations
- Write a short interpretation that connects summaries and plots to your insights

---

3.3 Textbook Connection

This assignment builds directly from **Chapter 3: Visualizations** in Reproducible Research Using R.

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

3.4 Submission Instructions

Submit **one `.R` script file**.

Your script must:

- Run from top to bottom without errors
- Include your insight statements and written summary as **comments** where requested
- Contain **three `ggplot2` plots** with clear titles and labeled axes
- Include at least **one faceted plot** using `facet_wrap()`

3.5 Assignment Tasks

3.5.1 Data Source

You must use the `nycOpenData` package.

3.5.2 Cleaning and Feature Engineering

Complete all of the following in your script:

3.5.2.1 Missing Data Removal (Defensible Choice)

- Choose **one column** where it is reasonable to remove missing values.
- Compute and report (in a comment) **how many values are missing** in that column before removal.
- Remove rows with missing values in that column.
- In a comment, explain **why** that column was "safe" to use.

3.5.2.2 Lowercase a Column

- Choose one categorical column **other than** `boro` (e.g., `perp_race`, `vic_race`) and convert all values to lowercase.

3.5.2.3 Create `time_of_day`

- Create a new column called `time_of_day` with three categories:
  - Morning
  - Afternoon
  - Night
- You define the time boundaries.
- Briefly note your boundaries in a comment.

3.5.2.4 Create `days_since`

- Create a column called `days_since` showing the number of days between today's date and the date of the shooting.
- Use `as.Date()` and/or `lubridate` as needed.

3.5.2.5 Turn Borough Names to Title Case

- In the `boro` column, convert each borough name to title case, so that only the first letter of each word is capitalized.
  - For example: `BROOKLYN` → `Brooklyn`

3.5.3 Insights

Provide the following insights as **comments** in your script:

1. **One insight about `time_of_day`**
   Example format: "Shootings are more frequent during _____ than _____."
2. **One additional insight of your choice**
   Explore anything you find interesting in the dataset. Be creative!

3.5.4 Visualizations (`ggplot2`)

Create **three visualizations** using `ggplot2`:

1. **Time of Day Plot:** Create a visualization that directly relates to your `time_of_day` variable.
2. **Personal Insight Plot:** Create a visualization that supports the **second insight you identified** in the previous section.
3. **Your Choice Plot:** Create a visualization that explores another relationship, pattern, or trend in the data that you find interesting.

For each visualization:

- Choose an appropriate `geom_*()` for the data you are visualizing
- Include an informative title
- Include clear x- and y-axis labels
- Use color through an aesthetic mapping (`aes(color = ...)` or `aes(fill = ...)`)
- Customize font and/or text sizes using `theme()`

Across your three visualizations:

- Use **at least two different `geom_*()` types**
- At least **one** visualization must use `facet_wrap()`

3.5.5 Written Summary

At the end of your script, write a short paragraph (as comments) describing:

- What you discovered from your insights
- What the graphs revealed that was not obvious from the raw data

---

3.6 Reproducibility Practice

This assignment focuses on reproducibility when working with **real civic data** and **date/time feature engineering**.

In your script:

- Clearly document your analytic choices (especially your `time_of_day` boundaries) in comments.
- Keep your pipeline modular by creating a cleaned object (e.g., `cleaned`), and build from it.
- After creating your new variables, include a quick checkpoint such as:
  - `count(cleaned, time_of_day)` or
  - `summary(cleaned$days_since)` to verify your features were created correctly.

The goal is that another reader can see *exactly what choices you made*, reproduce your engineered variables, and understand how you arrived at your insights.

4 Merging Data and Comparing Means with t-tests

4.1 Overview

This assignment builds on our lesson about **merging datasets**, **handling missing data**, and running **t-tests** in R. You will work with two files:

- **Employee Info** (Excel): first name, last name, department (30 employees)
- **Employee Scores** (CSV): a single `Name` column plus two test scores (`Score Before`, `Score After`)

Your goal is to merge the datasets, clean them, and answer questions using **t-tests**.

4.2 Learning Objectives

By completing this assignment, you will be able to:

- Import data from both Excel and CSV formats into R
- Resolve key mismatches across datasets (split/unite names to form a join key)
- Merge datasets using a defensible join strategy
- Diagnose and handle missing values in numeric columns
- Run and interpret:
  - a **paired (dependent) t-test**
  - an **independent t-test**
- Create a derived variable (`Improvement`) and use it in statistical comparisons
- Communicate results clearly (test statistic, p-value, interpretation)

4.3 Textbook Connection

This assignment builds directly from **Chapter 4: Comparing Two Groups: Data Wrangling, Visualization, and t-Tests** in Reproducible Research Using R.

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

4.4 Submission Instructions

Submit **one `.R` script file**.

Your script must:

- Run from top to bottom without errors
- Include clear section comments separating each part of the assignment
- Include your interpretations and written summary as **comments** where requested
- Load all packages used via `library(...)`

4.5 Assignment Tasks

4.5.1 1. Data Wrangling (Import + Merge)

1. Import both datasets into R:
    - Employee Info (Excel)
    - Employee Scores (CSV)
2. Notice the mismatch:
    - one dataset has `First Name` + `Last Name`
    - the other has a single `Name` column
3. Create a matching join key by doing **one** of the following:
    - **Option A:** split `Name` into first/last, or
    - **Option B:** unite first/last into a single full name

- **Clean the names:** make sure all column names are lowercase and are connected through an _ (e.g. score_before)

> When cleaning column names:
>
> The `clean_names()` functio from the `janitor` can be very helpful.

4. Rename columns if needed so the join key matches in both datasets.
5. Merge the datasets so each employee has:
    - department
    - first name and last name (or full name)
    - Score Before and Score After
6. After merging, you should have **30 rows**.
    - Write a short comment confirming the row count and what join you used.

4.5.2 2. Handling Missing Data (Imputation)

Some `Score Before` or `Score After` values have been removed on purpose.

1. Calculate the average of each score column:
    - `Score Before`
    - `Score After`
2. Replace missing values in each column with that column's average.
    - *Tip:* `mean(..., na.rm = TRUE)` and `replace_na()` (or base R) will be useful.
3. Write a comment confirming how many missing values were replaced in each column.

4.5.3 3. Dependent t-test (Paired)

1. Run a **paired t-test** comparing `Score Before` vs `Score After`.
2. Report (as comments):
    - test statistic
    - p-value
    - a 1–2 sentence interpretation (Did performance change after the intervention?)

4.5.4 4. Independent t-tests (Sex + Department)

4.5.4.1 4A. Add a Sex column (using `babynames`)

1. Add a `Sex` column by merging on first names using the `babynames` package.
2. If a name does not have an associated sex, remove it from the dataset.
3. Write a short comment reporting how many rows remain after this filtering step.

4.5.4.2 4B. Improvement t-test by Sex

1. Create a new column:

```
Improvement = score_after - score_before
```

2. Run an **independent t-test** comparing `Improvement` between male and female employees.
3. Report (as comments):
    - test statistic
    - p-value
    - interpretation

4.5.4.3 4C. Improvement analysis by Department

Using the same `Improvement` column, run a t-test in conjunction with the `department` column.

You can do this in one of two ways:

- If department has **two groups**, run a standard independent t-test:
  - `t.test(Improvement ~ department, data = ...)`
- If department has **more than two groups**, run **multiple** pairwise t-tests (or clearly explain why a t-test is not appropriate and what you would use instead).

Write a few sentences summarizing what you found.

4.5.5 Written Summary

At the end of your script, write a short paragraph (as comments) describing:

- Your overall findings across the t-tests
- Any difficulties you experienced during merging, missing data handling, or interpretation

---

4.6 Reproducibility Practice

This assignment focuses on reproducible workflows for **joining datasets** and **handling missing data**, where mistakes can silently change results.

In your script:

- Print or comment the number of rows **before and after** the merge (confirm you ended with 30 rows).
- Explicitly document which join you used (e.g., `left_join`) and *why*.
- After replacing missing values, include a "checkpoint" that confirms there are no remaining `NA`s in the score columns (e.g., `sum(is.na(score_before))`).
- Keep key variables named consistently (e.g., `score_before`, `score_after`, `improvement`) so your analysis is easy to verify.

The goal is that another reader could review your script and confirm the merge logic, the imputation logic, and the statistical results without guessing.

5 R Script → Quarto Report

5.1 Overview

This week, you will take your Week 3 homework assignment (NYPD Shooting Incidents) and convert it into a **Quarto** report (`.qmd`). Using the same prompt and the work you already completed, your task is to create a fully reproducible, publishable HTML report.

Your final output should read like a short research note: code + narrative explanations + tables/plots + brief interpretation.

5.2 Learning Objectives

By completing this assignment, you will be able to:

- Convert an analysis workflow from an R script into a reproducible Quarto document
- Organize work into labeled code chunks that render cleanly to HTML
- Combine narrative text with analysis code to communicate findings
- Use inline R expressions to report values dynamically
- Create publication-quality tables and visualizations for the web
- Publish a reproducible report and share it via a link

5.3 Textbook Connection

This assignment builds directly from **Chapter 10: Reproducible Reporting** in Reproducible Research Using R.

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

5.4 Submission Instructions

Submit **two things**:

1. A link to your published HTML report (e.g., RPubs, Posit Connect/Cloud, GitHub Pages, etc.)
2. Your `.qmd` file

Your Quarto report must render to HTML without errors.

5.5 Assignment Tasks

Your Quarto document must include the following sections and components.

5.5.1 1. YAML header

Include:

- Title
- Your name as the *author*

5.5.2 2. Setup chunk

Load all required libraries (for example: `tidyverse`, `lubridate`, `ggplot2`, `nycOpenData`, etc.).

5.5.3 3. Minimum required code chunks (4)

Your report must contain *at least* the following four code chunks:

1. **Data ingest**
   Use `nycOpenData::nyc_shooting_incidents()` to load the NYPD shootings dataset.
2. **Data cleaning**
   At least **two** cleaning/feature steps (examples: dropping NAs, creating `time_of_day`, lowercasing a column, creating `days_since`).

3. **Insights**
   Include the code used to generate your Week 3 insights (grouped summaries, counts, etc.).

4. **Tables & Graphs**

   - At least **one table** created with `knitr::kable()`
   - At least **two ggplot2 graphs**:
     - one related to `time_of_day`
     - one supporting your personal insight

All tables and graphs must have captions!

5.5.4 4. Narrative text

After **each** code chunk, write a short explanation describing:

- what you did
- why you did it
- what you learned or produced

5.5.5 5. Inline R expression

Include at least **one** inline R expression in your narrative text.

Example:

> The dataset contains `r nrow(shooting_data)` rows after cleaning.

5.5.6 6. Hyperlink to the dataset

Provide a clickable link to the dataset (not a raw URL pasted as plain text).

Example:

NYC Open Data Portal — NYPD Shooting Incident Data

(You may link directly to the dataset page if you prefer.)

5.5.7 Reflection

In **3–5 sentences**, explain how this workflow could potentially help you in your thesis research.

5.6 Publishing Instructions

1. Render your Quarto document to HTML.
2. Publish it online (e.g., RPubs).
3. Submit:
    - the link to the published HTML
    - your `.qmd` file

5.7 Reproducibility Practice (Quarto Reporting)

This assignment focuses on reproducibility as **communication**.

In your report:

- Chunk labels should be meaningful (e.g., `data-ingest`, `cleaning`, `insights`, `plots`)
- Your narrative should match your code (a reader should understand what each chunk accomplishes)
- Inline R values should be used to avoid hard-coding numbers that may change if the dataset updates
- Your HTML output should render cleanly and be readable without needing the original script

6 ANOVA and Payment Patterns in NYC Camera Violations

6.1 Overview

You are working as a data scientist hired by a law firm that specializes in fighting parking and camera tickets.

The firm wants to understand whether certain groups tend to pay more than others. Previously, we explored patterns by day of week and time of day. Now, you will conduct a **structured ANOVA investigation** using NYC camera violation data.

Your job is to determine whether payment differences across groups are:

- Statistically significant
- Practically meaningful
- Strategically useful for marketing

Be sure to include a hyperlink somewhere in your introduction to the dataset via the NYC Open Data Portal.

6.2 Learning Objectives

By completing this assignment, you will be able to:

- Load public NYC data using an R package
- Clean and prepare real-world categorical and numeric variables
- Engineer time-based features (day of week, time of day)
- Conduct multiple one-way ANOVA tests
- Interpret F-values, p-values, and PRE
- Distinguish between statistical and practical significance

- Communicate inferential findings clearly in a reproducible report

---

6.3 Textbook Connection

This assignment builds directly from **Chapter 5: Comparing Multiple Groups** in Reproducible Research Using R.

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

6.4 Submission Instructions

Submit **two files**:

1. Your `.qmd` file
2. Your knitted `.html` file

Your document must:

- Knit without errors
- Contain clearly labeled sections
- Include visualizations, descriptive statistics, and ANOVA results
- Include written interpretation after each major section
- Use at least one inline R expression

6.5 Assignment Tasks

You must complete the following sections in order.

6.5.1 1. Data Ingest

Load the NYC camera violations dataset using the `nycOpenData` package.

Your section must:

- Clearly state what the dataset represents
- Report how many rows are in the dataset (use an inline R expression)
- Confirm that the dataset loaded successfully

Do not hard-code numbers in your text.

6.5.2 2. Data Cleaning and Feature Engineering

You must perform the following cleaning steps. You are responsible for determining how to implement them.

6.5.2.1 2A. Numeric Conversion

Ensure all payment-related variables (such as `payment_amount` and `fine_amount`) are stored as numeric.

Include a short checkpoint confirming the structure is correct.

6.5.2.2 2B. Date Cleaning

The `issue_date` column may contain inconsistent formats.

You must:

- Retain only properly formatted dates
- Convert `issue_date` into a Date object
- Create a new variable called `day_of_week`

Explain briefly why cleaning dates is necessary before modeling.

6.5.2.3 2C. Time Cleaning

Using the `violation_time` column:

- Convert it into a format that can be interpreted numerically
- Create a new categorical variable called `time_of_day`

Your categories must include:

- Morning
- Afternoon
- Night

Include a checkpoint showing how many observations fall into each category.

Explain why feature engineering improves interpretability.

6.5.3 3. Analysis 1: Day of Week and Fine Amount

You will now investigate whether average **fine_amount** differs by **day_of_week**.

Your section must include:

- A grouped descriptive summary (means displayed in descending order)
- A one-way ANOVA
- A `supernova()` table
- A post-hoc test (Tukey)

Then write a short paragraph answering:

- Is the effect statistically significant?
- How much variance is explained (PRE)?

- Is the effect practically meaningful?
- Why might statistical significance be misleading here?

6.5.4 4. Analysis 2: Time of Day and Fine Amount

Repeat the same structure, but now examine **fine_amount by time_of_day**.

Include:

- Visualization (boxplot required)
- Descriptive statistics
- ANOVA
- PRE interpretation

In your paragraph, explain:

- Whether differences are meaningful
- Why fine amounts may show significance even if policy does not change

---

6.5.5 5. Analysis 3: Violation Type and Fine Amount

Examine **fine_amount by violation type**.

Your section must include:

- Descriptive statistics
- ANOVA
- PRE interpretation

In your paragraph, explain:

- Why this model is statistically strong
- Why it may not be conceptually interesting
- What this teaches us about modeling fixed policy variables

6.5.6 6. Analysis 4: Violation Type and Payment Amount

Now shift to the more behaviorally meaningful variable: **payment_amount**.

Examine **payment_amount by violation type**.

Your section must include:

- Visualization (boxplot required)
- Descriptive statistics
- ANOVA
- PRE interpretation

In your paragraph, explain:

- Why this analysis is more practically meaningful than fine_amount
- What the size of PRE suggests
- How a law firm might use this insight strategically

6.5.7 Final Reflection

In 5–7 sentences, answer:

1. What is the difference between statistical significance and practical significance?
2. Which of the four analyses above do you think is most useful for real-world decision-making?

3. If you were advising the law firm, what is one concrete recommendation you would make based on your findings?

6.6 Publishing Instructions

1. Render your `.qmd` file to HTML.
2. Confirm that warnings and messages do not appear in the final document.
3. Submit both your `.qmd` file and your published `.html` report.

6.7 Reproducibility Practice

This assignment focuses on reproducible statistical modeling.

Your document must:

- Use clearly labeled code chunks
- Avoid hard-coding values in your narrative
- Use inline R expressions where appropriate
- Include at least one data-checking "checkpoint" in each major section
- Render cleanly to HTML without manual intervention

The goal is that another analyst could rerun your entire workflow and verify every statistical conclusion.

6.8 Citing Your R Packages

An important part of reproducible research is acknowledging the software that makes our analyses possible.

During today's group activity, we learned how to generate a bibliography and cite R packages directly in Quarto.

Now it's your turn to apply those skills independently!

6.8.1 Your Task

1. Create a file called `packages.bib` containing references for the R packages used in your assignment.
2. Save your bibliography file within your R project, either alongside your `.qmd` document or in a dedicated folder.
3. Update your Quarto YAML to include the bibliography.
4. Include citations for all R packages used in your analysis throughout your written report.
5. Make sure your references appear automatically when you render your document to HTML.

Your citations should appear naturally in your writing. For example, you might cite the package used to retrieve the dataset, the package used to create your visualizations, or the package used to conduct your statistical analyses.

> Need a Reminder?
>
> - `knitr::write_bib()` to generate a bibliography
> - `citation()` to explore how R packages should be cited
> - Adding `bibliography: packages.bib` to our Quarto YAML
> - Using citation keys such as `[@R-ggplot2]` within our writing
>
> Remember to check the citation keys inside your own `packages.bib` file!

Important: Your citations should be integrated into your report, not simply listed at the bottom. The goal is to acknowledge the tools you used while explaining your analytical process.

7 Midterm: Exercise & Sleep

7.1 Overview

You are a researcher in sleep expert Dr. **Matthew Walker's lab**. You have just run an experiment measuring the effect of different **exercise methods** on sleep habits.

You measured:

- Average hours of sleep **before** the experiment
- Average hours of sleep **after** the experiment
- **Sleep efficiency** at the end of the experiment

Dr. Walker needs you to identify **which (if any) exercise method best improves sleep**, and to make a clear, evidence-based recommendation.

This midterm assesses your ability to build a **fully reproducible analysis** from raw data through statistical inference and final reporting.

7.2 Learning Objectives

By completing this midterm, you will be able to:

- Import data from a multi-sheet Excel workbook
- Clean messy labels and standardize categorical variables
- Merge datasets into a single analysis-ready table
- Engineer derived variables and handle missing data responsibly
- Produce descriptive statistics and professional tables
- Create clear, interpretable visualizations
- Conduct and interpret t-tests and ANOVAs
- Run and interpret Tukey post-hoc comparisons
- Synthesize results into an actionable recommendation
- Create a professional, reproducible report using Quarto
- Incorporate inline R code and cross-references into scientific writing

- Properly cite R packages used in your analysis

7.3 Textbook Connection

This assignment builds directly from **Chapters 1-6 (and 10)** in Reproducible Research Using R.

Students are encouraged to review these chapters before beginning this assignment, as they provide the conceptual foundation and reproducible workflow demonstrated here.

7.4 Submission Instructions

Submit **three files**:

1. Your `.qmd` file
2. Your rendered and published `.html` file
3. Your `packages.bib` bibliography file

Your report must:

- Render without errors
- Include short narrative text between code chunks
- Use clear section headers matching the numbered tasks below
- Suppress unnecessary warnings/messages in the final HTML
- Include readable plots and professionally formatted tables
- Follow all Quarto and reproducibility requirements outlined below

---

7.5 Quarto & Reproducibility Requirements

In addition to completing the statistical analyses, you are expected to produce a **professional, fully reproducible Quarto report**.

7.5.1 1. YAML & Document Organization

Your report must include:

- A descriptive title
- Your name as the author
- A date
- A clickable table of contents
- Numbered sections

You are responsible for modifying your YAML to include these features.

7.5.2 2. Code Chunk Labels

Every R code chunk must have a **unique, descriptive label**.

Your labels should clearly communicate the purpose of the code.

For example:

- `data-import`
- `data-cleaning`
- `descriptive-statistics`
- `ttest-sex`

Additionally:

- All figure-producing chunks must begin with `fig-`
- All table-producing chunks must begin with `tbl-`

7.5.3 3. Figures, Tables & Cross-References

All visualizations and tables must have descriptive captions and appropriate labels.

You must also **reference every figure and table in your written narrative** using Quarto's cross-referencing system.

For example:

- `@fig-sleep-difference`
- `@tbl-descriptive-statistics`

Your references should appear naturally in your interpretation, just as they would in a published research paper.

Simply referencing a figure or table is not enough. You must explain what the results suggest.

> Remember!
>
> Quarto automatically numbers your figures and tables when you use the appropriate labels and captions.
> This means you should not manually write "Figure 1" or "Table 2" in your interpretations.
> Instead, use cross-references so your document updates automatically!

7.5.4 4. Inline R Code

Your report must include **at least three instances of inline R code**.

At least one instance must appear in your final synthesis and recommendation.

Use inline R code to dynamically report statistical values, such as:

- Means
- Standard deviations
- Test statistics
- p-values

Do not manually type statistical results into your narrative when they can be generated using inline R code.

Your report should automatically update if the underlying data or analysis changes.

7.5.5 5. Citing Your R Packages

All R packages used in your analysis must be properly cited.

You must:

1. Create a file called `packages.bib` containing references for the R packages used in your analysis.
2. Include your bibliography file in your Quarto YAML.
3. Cite all R packages used throughout your written report.
4. Ensure your references appear automatically when you render your document to HTML.

Your citations should appear naturally in your writing.

Need a Reminder?

You can use `knitr::write_bib()` to generate bibliography entries for your R packages. Review our previous Quarto lesson and assignments if you need a refresher on creating a bibliography and using citations.

7.6 Midterm Tasks

7.6.1 1. Setup & Data Import

7.6.1.1 Tasks

- Load all packages you use.
- Import **both sheets** from `midterm_sleep_exercise.xlsx` using `readxl`:
  - `participant_info_midterm`
  - `sleep_data_midterm`
- Preview both datasets using appropriate functions (e.g., `glimpse()` or `head()`).

7.6.1.2 Requirements

- Use clearly labeled code chunks.
- Briefly describe the datasets you imported.
- Include an appropriate citation for the package used to import your data.

7.6.2 2. Merge & Base Cleaning

7.6.2.1 Tasks

- Clean column names so they are consistent and easy to work with.
- Standardize labels for:
  - `Exercise_Group` (e.g., "cardio", "CARDIO" → "Cardio")

  - `Sex` (e.g., "m", "MALE" → "Male")

- Merge the two datasets into a single dataset.

7.6.2.2 Requirements

- Final dataset should be **one row per participant**.
- Dataset should include demographics + sleep variables.
- Include a checkpoint verifying row count and that the merge behaved as expected.
- Briefly explain your cleaning and merging decisions.

7.6.3 3. Create Derived Variables

7.6.3.1 Tasks

- Create a numeric sleep-change variable:
  - `Sleep_Difference = post_sleep - pre_sleep_num`
- Create `AgeGroup2` using `case_when()` with exactly **two groups** (example: `< 40` vs `>= 40`).
- Identify how many rows are missing `Sleep_Difference`.
- Drop rows with missing `Sleep_Difference`.

7.6.3.2 Requirements

- Include a checkpoint showing how many rows were removed and why.
- Your narrative should briefly justify your missing-data decision.
- Use inline R code to report how many observations were removed.

7.6.4 4. Descriptive Statistics

7.6.4.1 Tasks

Report descriptive statistics for:

- `Sleep_Difference` (overall)
- `Sleep_Efficiency` (overall)

For each variable, include:

- Mean
- SD
- Min
- Max

Then report group-wise summaries by `Exercise_Group` for:

- `Sleep_Difference`
- `Sleep_Efficiency`

7.6.4.2 Requirements

- Group summaries must be clearly readable.
- Briefly interpret the most meaningful patterns in your descriptive statistics.
- Tables must be formatted professionally using `kable()` or an equivalent table-formatting function.
- All tables must include descriptive captions using Quarto's `tbl-cap` option.
- All table-producing chunks must use the `tbl-` prefix.
- **Reference every table in your written narrative** using Quarto cross-references.

7.6.5 5. Visualizations (3 Plots)

Create **three** plots:

1. Boxplot: `Sleep_Difference` by `Exercise_Group`
2. Boxplot: `Sleep_Efficiency` by `Exercise_Group`
3. Scatterplot: `Sleep_Difference` vs `Sleep_Efficiency`, with a trend line

7.6.5.1 Requirements

Each plot must include:

- Descriptive title
- Clear x- and y-axis labels
- A clean theme (e.g., `theme_minimal()`)
- A descriptive Quarto figure caption using `fig-cap`
- A unique chunk label beginning with `fig-`

After each plot, write **1–3 sentences** describing what the plot suggests.

You must **reference each visualization directly in your written interpretation** using Quarto's cross-referencing system (e.g., `@fig-sleep-difference`).

Your interpretation should discuss meaningful patterns, differences, or relationships shown in the visualization.

7.6.6 6. t-tests (Two)

Run **two** t-tests:

1. `Sleep_Difference ~ Sex`
2. `Sleep_Difference ~ AgeGroup2`

7.6.6.1 Requirements

For each t-test, report:

- Group means
- t statistic
- df
- p-value

Then interpret in plain language:

- Is the difference statistically significant?
- Is it practically meaningful?

7.6.7 7. ANOVAs (Two) + Post-hoc Tests

Run **two** ANOVAs:

- ANOVA A: `Sleep_Difference ~ Exercise_Group`
- ANOVA B: `Sleep_Efficiency ~ Exercise_Group`

7.6.7.1 Requirements

For each ANOVA, include:

- ANOVA table output
- F statistic
- df
- p-value
- PRE from `supernova()` (required)
- A brief interpretation of the effect size

Then run Tukey post-hoc tests:

- `TukeyHSD()` for each ANOVA

Interpret results:

- Which groups differ significantly?
- Which exercise group appears to "win" for each outcome?
- How much variation in the outcome is explained by exercise group, based on PRE?

7.6.8 8. Synthesis & Recommendation

Write **4–6 sentences** answering:

- Considering **both** outcomes (`Sleep_Difference` and `Sleep_Efficiency`) and your post-hoc results, which **single** exercise regimen would you recommend to improve overall sleep?

7.6.8.1 Requirements

- You may only pick **one** regimen.
- Support your recommendation with specific statistical evidence:
  - F values and p-values
  - Key Tukey contrasts
  - Any meaningful patterns from your plots/tables
- Include at least one instance of inline R code.
- Reference relevant figures and/or tables from earlier in your report using Quarto cross-references.
- Acknowledge any important limitations or conflicting evidence.

Your recommendation must be **actionable and executive-friendly**.

Imagine Dr. Walker only has time to read this section. He should understand your recommendation and the evidence supporting it without needing to examine your R code.

7.6.9 9. Reflection

Write **3–5 sentences** addressing:

- What was most challenging?
- What did you feel confident about?
- What would you do differently next time to improve your analysis or report?

You may reflect on your statistical analyses, data cleaning, Quarto workflow, or reproducibility practices.

7.7 Publishing Instructions

1. Render your `.qmd` file to HTML.
2. Confirm that your table of contents appears and works correctly.
3. Verify that all figures and tables are properly numbered and cross-referenced.
4. Confirm that inline R code displays the correct values.
5. Confirm that your package citations and references appear correctly.
6. Confirm that unnecessary warnings and messages do not appear in the final document.
7. Submit your `.qmd`, published `.html`, and `packages.bib` files.

7.8 Reproducibility Practice

This midterm evaluates not only statistical accuracy, but **workflow transparency and professional scientific communication**.

Your report must:

- Load all packages explicitly
- Import both sheets programmatically
- Confirm merge success with row counts
- Explicitly handle missing values
- Use unique, descriptive labels for every R code chunk
- Use `fig-` and `tbl-` prefixes for figures and tables
- Include captions and cross-references for every figure and table
- Incorporate inline R code into your written interpretations
- Cite all R packages used in the analysis
- Generate a references section automatically
- Avoid hard-coding statistical values in written interpretation
- Suppress unnecessary warnings/messages
- Render cleanly to HTML

Remember: Your goal is not simply to run statistical tests. Your goal is to produce a fully reproducible research report that another researcher could read, understand, and reproduce.

8 CUNY Analytics — Exploring Team Performance Through Reproducible Analysis

8.1 Overview

You have been hired as a **Sports Data Analyst for the CUNY Athletic Conference**. Your assignment is to investigate the relationships between basketball statistics and game outcomes for **Brooklyn College men's basketball**, using completed post-COVID seasons **2021–2022 through 2025–2026**.

Use the `CUNYsports` R package and the `basketball_game_by_game_team_overall_final` table. Each observation should represent one Brooklyn College game; verify this after filtering. Your report should be a polished, reproducible **Quarto chapter**. Next week's linear regression chapter will extend the question you develop here.

> **i** Dracula Theme
>
> For this assignment, you must use the **Dracula theme** throughout your Quarto document and all visualizations.
> You can find the official Dracula theme, color palette, and additional resources on GitHub.
> Remember that `highlight-style: dracula` only changes syntax highlighting. You must also apply Dracula styling to your HTML document and your `ggplot2` visualizations, as demonstrated in class.

8.2 Learning Objectives

By completing this assignment, you will:

- Retrieve data reproducibly from an R package.
- Filter and validate game-level data for a specific school, division, and set of seasons.
- Create and interpret descriptive tables and Dracula-themed visualizations.
- Conduct Pearson, point-biserial, and partial correlations.

- Explain statistical findings and limitations for a nontechnical audience.
- Cite software, use inline R, and cross-reference sections, figures, and tables in Quarto.
- Develop a question that will carry forward into next week's regression chapter.

8.3 Textbook Connection

Review **Chapter 7: Correlation** in *Reproducible Research Using R*, along with the section on **citing R packages**.

8.4 Submission and Formatting Requirements

Submit **three files**: your `.qmd` source, your packages.bib file, and the rendered `.html` link from posit connect cloud. Your report must:

1. Render successfully with no unnecessary warnings or messages.
2. Use a **Dracula-themed HTML document and Dracula-themed plots** (not only Dracula syntax highlighting).
3. Include descriptive writing, clearly labeled sections, and a professional executive memo.
4. Cite **every R package you use**, including `CUNYsports`, using the package citation workflow from the textbook. Include an appropriate references section/bibliography.
5. Include **at least one inline R expression** that reports a computed value.
6. Give **every section** a unique Quarto identifier in the form `{#sec-name}` and refer to **at least one section** using `@sec-name` in your prose.
7. Give **every table and figure** a Quarto label (`tbl-...` or `fig-...`) and caption. Refer to **at least two figures and two tables** in your written interpretation using cross-references such as `@fig-season-comparison` and `@tbl-season-by-season`. Do not merely list references: integrate them into complete sentences.
8. Avoid hard-coding numerical findings in your prose when an inline R expression would be appropriate.

> **Warning**
>
> A cross-reference works only if its target exists and is correctly labeled. For example, a table with chunk label `tbl-season-by-season` can be discussed as: "As shown in **?@tbl-season-by-season**, …".

8.5 Assignment Tasks

8.5.1 Retrieve and Explore the Data

Load `CUNYsports`, `tidyverse`, `ggcorrplot`, and any other packages you need. Use the `CUNYsports` documentation functions to inspect the table and variables, then retrieve:

`basketball_game_by_game_team_overall_final`

Before filtering, examine the unique values in `school`, `gender`, `season_label`, `home_away`, and `team_side`.

8.5.2 Prepare Brooklyn College Men's Basketball Data

Filter the data to:

- **School:** Brooklyn College.
- **Division:** Men's basketball.
- **Seasons:** 2021–2022, 2022–2023, 2023–2024, 2024–2025, and 2025–2026.

Confirm that your final data contains one observation per game and that each requested season is represented. If a season is unavailable, report that rather than inventing observations.

Create:

- `Home_game`: 1 for home games and 0 for away games. Exclude neutral-site games **from analyses using this binary variable**, but retain them for other analyses if appropriate.
- `STOCKS`: `steals + blocks`.

Check for missing values and confirm that `point_differential` is measured from Brooklyn College's perspective. Describe how you handled missing or invalid observations.

8.5.2.1 Table: Season-by-Season Summary [#sec-season-by-season-plot]

Create a **captioned, cross-referenceable table** labeled `tbl-season-by-season`, reporting for each season:

- Number of games.
- Wins and losses.
- Average point differential.

Use the `DT` package to display this data. Include a sentence interpreting this table and citing it using `@tbl-season-by-season`.

8.5.2.2 Table: Home and Away Summary

Create a second captioned table labeled `tbl-home-away` summarizing the number of games and mean point differential for home and away games. Reference it in your written discussion using `@tbl-home-away`.

8.5.3 Visual Exploration

Create **at least three Dracula-themed figures** using `ggplot2`. Each must have a meaningful title, readable axis labels, a caption, and a Quarto figure label.

8.5.3.1 Figure: Season-by-Season Performance

Create a **boxplot** showing `point_differential` by `season_label` across the five seasons. Label it `fig-season-comparison`.

In 3–4 sentences, discuss which seasons appear stronger or weaker, differences in variability, and any unusual games. Reference `@fig-season-comparison` in your interpretation.

8.5.3.2 Figure: Shooting Efficiency and Game Outcomes

Create a scatterplot with:

- X-axis: `effective_field_goal_percentage`.
- Y-axis: `point_differential`.
- Color: `home_away`.
- A linear trend line using `geom_smooth(method = "lm")`.

Label it `fig-shooting-margin`. Discuss the direction, apparent linearity, and possible outliers. Reference `@fig-shooting-margin` in your discussion.

8.5.3.3 Figure: Your Choice

Choose one additional relationship to visualize, such as turnovers versus point differential, assists versus scoring, or STOCKS versus point differential. Label it `fig-independent-exploration` and explain why you selected it.

All three figures must use the Dracula theme. A default `ggplot2` theme does not meet this requirement.

8.5.4 Correlation Analysis

8.5.4.1 Point-Biserial Correlations

Using the binary `Home_game` variable, examine its relationship with:

- `point_differential`.
- `effective_field_goal_percentage`.

Report r, direction, strength, statistical significance, and plain-language meaning for both. Explain whether the results provide evidence of a home-court association.

8.5.4.2 Correlation Matrix

Create a correlation matrix containing at least:

- `point_differential`
- `effective_field_goal_percentage`
- `total_rebounds`
- `assists`
- `turnovers`
- `steals`
- `blocks`

Visualize it with `ggcorrplot`, using Dracula styling. Label the result `fig-correlation-matrix` and cite `@fig-correlation-matrix` in your discussion.

Answer: Which variables show the strongest positive and negative associations with point differential? Which relationships surprised you?

8.5.4.3 Partial Correlation

First calculate the Pearson correlation between **assists** and **point differential**. Then calculate the partial correlation between the same two variables while controlling for **effective field-goal percentage**, using `ppcor`.

Explain what "controlling for" means, compare the two coefficients, and discuss whether the relationship changes. Use the same complete-case observations for both coefficients so the comparison is meaningful.

8.5.5 Your Independent Research Question

Develop **one research question** about Brooklyn College men's basketball across the five seasons. It must include a quantitative outcome and at least one quantitative explanatory variable.

State your question, explain its importance, identify the variables, create a scatterplot (you may reuse Figure 3 if appropriate), calculate a Pearson correlation, and interpret the result in one paragraph.

Next week's connection: You will use **the same dataset and research question** for a linear regression analysis. Choose a question you would like to explore further. End this section by explaining how regression might extend your correlation findings.

8.6 Executive Memo

Write a short professional memo addressed to:

CUNY Athletic Conference — Athletics Administration

Summarize the main findings, home-court associations, performance metrics most closely related to scoring margin, one limitation, and one recommendation for future analysis. Write for a nontechnical audience. Refer to at least one of your labeled tables or figures within the memo.

8.7 Reflection and Looking Ahead

In approximately 4–6 sentences, address:

1. How does `CUNYsports` support reproducible research?
2. What did the season-by-season analysis reveal that a single-season analysis might miss?
3. What did partial correlation add to your interpretation?
4. How might next week's regression chapter build on this work?

8.7.1 Reproducibility Practice

This assignment emphasizes **reproducible research, professional presentation, and effective scientific communication**.

Your Quarto document must:

- Retrieve the basketball data programmatically using the `CUNYsports` package.
- Filter the dataset to include Brooklyn College men's basketball across the five post-COVID seasons (2021–2022 through 2025–2026).
- Avoid manual data entry or modification.
- Use clearly labeled R code chunks.
- Avoid hard-coding statistical results in written interpretations; use inline R expressions where appropriate.
- Apply the **Dracula theme** to your Quarto document and all visualizations.
- Include descriptive titles, axis labels, and captions for all figures and tables.
- Use Quarto cross-references for figures (`@fig-name`), tables (`@tbl-name`), and sections (`@sec-name`) at least once each in your written discussion.
- Assign a unique section identifier using `{#sec-name}` to every major section of your document.
- Properly cite **all R packages used** in your analysis, following the textbook's instructions.
- Render cleanly to HTML without unnecessary warnings, messages, or errors.

Can we build upon this…?

This assignment will serve as the foundation for **next week's linear regression chapter**. Organize your analyses, research questions, and interpretations so that you can

build upon them in your next assignment.

The goal is that another researcher could reproduce your entire analysis, understand your analytical decisions, and follow your findings without needing additional instructions. Your final product should be something you would be proud to include in a publicly accessible research portfolio.

8.8 Final Checklist

- ☐ Filtered to Brooklyn College men's basketball for the five specified seasons.
- ☐ Verified team-side records, game uniqueness, and category coding.
- ☐ Included `tbl-season-by-season` and `tbl-home-away` with captions and prose references.
- ☐ Included season boxplot, shooting-efficiency scatterplot, and independent visualization.
- ☐ Included a Dracula-themed correlation matrix.
- ☐ Used Dracula styling for **the entire HTML report and every plot**.
- ☐ Completed point-biserial, Pearson, and partial correlations.
- ☐ Developed a question to carry into next week's regression assignment.
- ☐ Cited all R packages and included a bibliography.
- ☐ Used an inline R expression.
- ☐ Gave every section a `{#sec-name}` ID and used at least one `@sec-name` reference.
- ☐ Used cross-references to at least two figures and two tables in prose.
- ☐ Submitted the `.qmd`, `.bib`, and posit connect cloud link.

9 CUNY Analytics — From Correlation to Regression

9.1 Overview

Last week, you worked as a **Sports Data Analyst for the CUNY Athletic Conference**, investigating relationships between basketball statistics and game outcomes for Brooklyn College men's basketball across the 2021–2022 through 2025–2026 seasons.

You prepared the data, explored team performance, conducted correlation analyses, and developed an independent research question. **This week, you will continue that same investigation—not start a new one.**

Your new assignment is to use **linear regression** to examine how well your original explanatory variable accounts for differences in your chosen quantitative outcome, whether adding other basketball statistics improves the model, and whether the conclusions you drew last week still hold.

Your final product should be a **fully reproducible Quarto chapter** that can eventually follow your correlation chapter in your Quarto book.

9.2 Learning Objectives

By completing this assignment, you will be able to:

- Extend an existing research question from correlation to regression.
- Fit and interpret a simple linear regression model.
- Explain a regression equation, predicted values, residuals, and R^2.
- Visualize fitted values and examine model assumptions.
- Fit and compare multiple regression models using R^2, Adjusted R^2, and AIC.
- Evaluate whether accounting for additional predictors changes your original conclusions.
- Communicate findings clearly in a reproducible, published Quarto report.

9.3 Textbook Connection

Review **Chapter 8: Linear Regression** in *Reproducible Research Using R*. You may also revisit **Chapter 7: Correlation** when connecting this week's findings to your previous chapter.

9.4 Submission and Formatting Requirements

Submit **three items**:

1. **Your `.qmd` source file** containing the complete analysis and writing.
2. **Your `packages.bib` file** containing citations for all R packages used.
3. **A working link to the HTML report published on Posit Connect Cloud.** A local `.html` file does **not** replace this link.

Your report must:

- Render without unnecessary warnings, messages, or errors.
- Use a consistent, professional visual style. **The Dracula theme is not required.**
- Include labeled and captioned figures and tables.
- Include at least **two figure cross-references**, **two table cross-references**, and **one section cross-reference** in your prose.
- Give every major section a unique Quarto identifier, such as `{#sec-simple-regression}`.
- Include at least one **inline R expression** reporting a calculated result rather than a hard-coded number.
- Cite all R packages used and include a references section using `packages.bib`.
- Include a professional executive memo.

> **i** Reduce, Reuse, Recycle!
>
> Continue using the **same prepared Brooklyn College men's basketball dataset and independent research question** from last week. Reuse your existing data preparation code as needed so this chapter renders reproducibly, but **do not repeat** the descriptive tables, exploratory plots, correlation matrix, or partial-correlation analysis from your previous report.

9.5 Assignment Tasks

9.5.1 Continuing Your Investigation

Begin by restating the independent research question you developed last week. Identify your **quantitative outcome variable** and your **primary quantitative explanatory variable**.

In **3–4 sentences**, summarize what your previous correlation analysis found (including its direction and strength) and explain what you hope regression will add. Refer to your previous chapter as the motivation for this new analysis; you do not need to calculate the correlation again.

If your original question cannot reasonably be addressed using linear regression, make a small, clearly explained adjustment while keeping the same underlying basketball topic and dataset.

9.5.2 Simple Linear Regression

Use `lm()` to fit **one simple linear regression model** that addresses your research question.

For example, if you investigated shooting efficiency and point differential:

```
model_1 <- lm(
  point_differential ~ effective_field_goal_percentage,
  data = basketball_data
)

summary(model_1)
```

The code above is an example. Use your own research question, variable names, and prepared dataset.

Required tasks:

1. Display your regression results and create a **coefficient table** using `broom::tidy()` or an equivalent method.
2. Identify and interpret the **intercept** and **slope**, including the units of the predictor and outcome.
3. Write your estimated regression equation in the form **predicted y = intercept + slope * x**.
4. Report the predictor's **p-value** and the model's **R^2**. Explain what R^2 means in the context of your research question.

5. Create **one scatterplot** with the fitted regression line using `geom_smooth(method = "lm")`. Give the figure a caption and refer to it in your discussion.

In a short paragraph, explain what regression tells you that your original correlation coefficient did not.

9.5.3 Predictions, Residuals, and Model Diagnostics

A regression model does more than describe a line: it generates predicted values, and the differences between those predictions and the actual outcomes are called **residuals**.

Required tasks:

1. Use `predict()` to calculate the model's fitted values and `residuals()` to calculate residuals. Add both to your analysis dataset, keeping the rows aligned with those used to fit the model.
2. Create a **residual plot** with predicted values on the x-axis, residuals on the y-axis, and a horizontal line at zero.
3. Conduct a **Breusch–Pagan test** using `lmtest::bptest()`.
4. Interpret the residual plot and test result. Do the residuals show a pattern? Is there evidence that residual variance changes across fitted values?

For example:

```
predict(model_1)
residuals(model_1)
lmtest::bptest(model_1)
```

In approximately **3–4 sentences**, explain what your diagnostics suggest about the model and identify one limitation. A nonsignificant Breusch–Pagan test does not prove that all regression assumptions are satisfied.

> Connection to Our Pizza Sales Example
>
> In class, we used `lm()`, `predict()`, and `residuals()` to investigate whether pizza prices could explain sales volume. You are applying the **same workflow** to basketball performance. Focus on interpreting the output rather than simply displaying it.

9.5.4 Multiple Regression and Model Comparison

Your simple model uses one explanatory variable. Now investigate whether including other basketball statistics improves how well your model explains variation in your outcome.

Build **four models in total**:

1. **Model 1 — Baseline:** Your simple regression model from the previous section.
2. **Model 2 — Expanded:** Your primary predictor plus one additional basketball statistic.
3. **Model 3 — Alternative:** Your primary predictor plus a different additional basketball statistic.
4. **Model 4 — Full:** Your primary predictor and both additional statistics.

For example:

```
m1 <- lm(point_differential ~ effective_field_goal_percentage,
         data = basketball_data)

m2 <- lm(point_differential ~ effective_field_goal_percentage + turnovers,
         data = basketball_data)

m3 <- lm(point_differential ~ effective_field_goal_percentage + total_rebounds,
         data = basketball_data)

m4 <- lm(point_differential ~ effective_field_goal_percentage +
           turnovers + total_rebounds,
         data = basketball_data)
```

Choose additional variables that make sense for your own research question. Fit all four models to the ***same complete-case observations*** *so their metrics can be compared fairly.*

Required tasks:

- Create a **model comparison table** reporting **R^2, Adjusted R^2, AIC, and the overall model p-value** for each model.
- Use `broom::glance()` and, if helpful, `purrr::map_df()` or `purrr::map_dfr()` to assemble the table.
- Examine the coefficients and statistical significance of the predictors in your full model using `summary()` or `broom::tidy()`.

In your interpretation, explain:

- Which model has the highest R^2 and Adjusted R^2.
- Which model has the lowest AIC.

- Whether additional predictors improved model fit enough to justify the added complexity.
- Whether your original predictor remains statistically significant after accounting for the other variables.
- Which model you would recommend and **why**.

Explain why **Adjusted R^2 and AIC** are useful when comparing models with different numbers of predictors. Remember that stronger fit to the observed games does not necessarily establish accurate prediction of future games.

Optional bonus — Interaction: Fit one additional model containing an interaction term between two predictors. Explain what the interaction represents and whether it changes your interpretation.

9.5.5 Did Regression Change Your Conclusions?

This section is the **bridge between your two research chapters**.

Return to the findings you reported in your correlation chapter and answer the following in **one thoughtful paragraph**:

> After conducting simple and multiple linear regression, do your original conclusions still hold? Did your original explanatory variable remain important after accounting for other basketball statistics? Did any new predictors change how you understand your research question? What did regression reveal that correlation alone could not?

Use evidence from your regression coefficients, statistical significance, and model comparison metrics. Reference at least one figure or table from this chapter.

Do not claim that an observed association proves that changing a statistic will cause a team to win more games.

9.6 Executive Memo

Write a short, follow-up professional memo addressed to:

CUNY Athletic Conference — Athletics Administration

Your memo must include:

- Your original research question and the most important new finding.
- Your recommended model and why you selected it.
- The most relevant predictor(s) and the model's R^2.

- A plain-language explanation of what the findings mean for basketball performance.
- **One limitation** of the analysis.
- **One actionable recommendation** for future analysis, coaching strategy, or performance evaluation.

Reference at least one labeled figure or table in your memo. Write for a coach or administrator who may not be familiar with statistical terminology.

9.7 Reflection

In **4–6 sentences**, answer:

1. How did regression extend your previous correlation analysis?
2. What did you learn from examining predicted values and residuals?
3. How did comparing multiple models affect your conclusions?
4. How will the connection between these two assignments improve your eventual Quarto book?

9.8 Publishing and Reproducibility

Your new chapter must be able to render independently and reproduce its results. **Reuse the necessary data preparation code from last week** rather than manually copying results or importing a previously rendered table. Do not recreate last week's descriptive analysis unless a small piece is genuinely necessary for the new model.

Before submitting:

1. Render your `.qmd` successfully to HTML.
2. Confirm that all captions, cross-references, inline R values, citations, and tables work correctly.
3. Check that warnings and unnecessary messages are suppressed in the published report.
4. Publish the HTML report to **Posit Connect Cloud**.
5. Open the **published URL** and confirm that the complete report displays correctly.
6. Submit your **`.qmd` file**, **`packages.bib` file**, and **working Posit Connect Cloud URL**.

> Two Homeworks, One Research Story
>
> Your previous homework asked: **What relationships exist?**
> This chapter asks: **How can we model those relationships, and what changes when we consider multiple variables?**

9.9 Final Checklist

- ☐ Continued the same basketball research question and reused existing data preparation code.
- ☐ Explained the connection to last week's correlation findings without repeating the earlier analyses.
- ☐ Fitted and interpreted a simple linear regression model.
- ☐ Included a coefficient table and regression scatterplot.
- ☐ Interpreted the slope, intercept, p-value, equation, and R^2.
- ☐ Generated fitted values and residuals.
- ☐ Included a residual plot and interpreted a Breusch–Pagan test.
- ☐ Fitted four regression models using comparable observations.
- ☐ Included a model comparison table with R^2, Adjusted R^2, AIC, and overall model p-values.
- ☐ Explained whether regression changed the conclusions from the previous chapter.
- ☐ Completed the executive memo and reflection.
- ☐ Included at least two figure, two table, and one section cross-references.
- ☐ Included an inline R expression, package citations, and a bibliography.
- ☐ Used a professional, consistent style (no required Dracula theme).
- ☐ Submitted the `.qmd`, `packages.bib`, and **published Posit Connect Cloud link**.

10 Streaming Analytics — Understanding Platform Popularity Across Age Groups

This homework represents an individual assignment, that has now been replaced with Section 9.1.

10.1 Overview

The **Streaming Analytics Division (SAD)** has hired you as their new **Data Analyst**. Your mission is to determine whether **age group influences streaming platform preference**.

The company wants to know whether certain platforms — Netflix, Hulu, Disney+, Amazon, or Other — appeal more strongly to specific age demographics (18–25, 26–40, 41+). Your analysis will guide targeted marketing strategies, promotional campaigns, and content investment decisions.

Using simulated survey data, you will conduct a **fully reproducible Quarto analysis** to determine whether **Age Category and Platform Preference are statistically related**.

You will:

- Construct and interpret contingency tables
- Perform a Chi-Square test of independence
- Examine observed, expected, and residual values
- Decompose the Chi-Square statistic into contributions
- Visualize contributions using a heatmap
- Compute and interpret Cramer's V
- Communicate findings clearly in plain language

10.2 Learning Objectives

By completing this assignment, you will be able to:

- Clean and summarize categorical data
- Construct contingency tables
- Conduct and interpret a Chi-Square test of independence
- Interpret observed, expected, and residual frequencies
- Decompose the Chi-Square statistic into cell-level contributions
- Visualize categorical data using stacked and clustered bar charts
- Calculate and interpret Cramer's V
- Communicate statistical findings in accessible language

10.3 Textbook Connection

This assignment builds directly from **Chapter 6: Analyzing Categorical Data** in Reproducible Research Using R.

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

10.4 Submission Instructions

Submit **two files**:

1. Your `.qmd` file
2. Your knitted and published `.html` file

Your document must:

- Knit without errors
- Suppress warnings and messages in the final HTML
- Include clearly labeled sections
- Include all required tables and visualizations
- Include full statistical interpretations
- Use at least one inline R expression

10.5 Assignment Tasks

10.5.1 1. Data Preparation

You have been provided with survey data including:

- Age Category (18–25, 26–40, 41+)
- Preferred Streaming Platform (Netflix, Hulu, Disney+, Amazon, Other)

10.5.1.1 Required Tasks

You must:

- Inspect the dataset
- Compute total counts for Age Category
- Compute total counts for Platform Preference
- Construct a contingency table showing the joint distribution of Age Category and Platform Preference

Briefly describe what the contingency table represents and why it is useful for categorical analysis.

10.5.2 2. Visualization

Create **two visualizations** using `ggplot2`.

10.5.2.1 Required Visualization 1

A **stacked bar chart** showing the proportions of platform preferences within each age group.

10.5.2.2 Required Visualization 2

A **clustered (side-by-side) bar chart** showing counts for each platform across age groups.

Both visualizations must:

- Include descriptive titles
- Include labeled axes
- Include a legend
- Use a professional or themed style (e.g., from `ggthemes`)

After each plot, provide 3–4 sentences interpreting what you observe.

10.5.3 3. Chi-Square Test of Independence

Conduct a **Chi-Square test of independence** to determine whether Age Category and Platform Preference are related.

In your write-up, report and interpret:

- The Chi-Square statistic (^{2})
- Degrees of freedom (df)
- The p-value

Then answer:

- Is there a statistically significant relationship?
- What does this result mean in plain language?

10.5.4 4. Observed, Expected, and Residual Values

From your Chi-Square test, extract:

- Observed counts
- Expected counts
- Residuals

Discuss:

- Which age–platform combinations show more people than expected

- Which show fewer people than expected
- Which residuals appear largest in magnitude

Explain what residuals tell us about patterns in the data.

10.5.5 5. Contributions to the Chi-Square Statistic

Break down the Chi-Square statistic into **individual cell contributions**.

Then:

- Calculate the percentage contribution of each cell to the total ^{2} value
- Identify which age–platform combinations contribute most to the overall result

Visualize these percentage contributions using a **heatmap**.

In your interpretation, explain:

- Which cells drive the overall Chi-Square result
- What this suggests about viewing habits across age groups

10.5.6 6. Effect Size (Cramer's V)

Compute **Cramer's V** to measure the strength of association between Age Category and Platform Preference.

Interpret the value using these guidelines:

- 0.00–0.10 → Very Weak
- 0.10–0.30 → Weak
- 0.30–0.50 → Moderate
- Above 0.50 → Strong

Provide a short, plain-language explanation of the effect size.

10.5.7 7. Final Interpretation

Write a professional summary addressing:

- Whether the relationship between Age and Platform is statistically significant
- Which age groups and platforms are driving the relationship
- The strength of the association (Cramer's V)
- What these findings mean in a real-world marketing context

Your summary should resemble a professional analytics report, not a textbook answer.

10.6 Deliverables

Your final report must include:

- The contingency table
- Both visualizations (stacked and clustered)
- Chi-Square results (², df, p-value)
- Observed, expected, and residual tables
- Contribution table and heatmap
- Cramer's V calculation and interpretation
- Final written summary

10.7 Guidelines for Interpretation

When discussing your results, always include:

- **Statistical significance** — Is the relationship real?
- **Direction of pattern** — Which combinations are over- or under-represented?
- **Strength of association** — Using Cramer's V
- **Plain-language meaning** — What does this imply about streaming habits?

Avoid jargon without explanation.

10.8 Reflection

In 4–6 sentences, answer:

1. Why is the Chi-Square test appropriate for categorical data?
2. What is the difference between statistical significance and effect size?
3. Why is it important to examine residuals and contributions rather than only the p-value?

10.9 Publishing Instructions

1. Render your `.qmd` file to HTML.
2. Confirm warnings and messages do not appear in the final document.
3. Submit both your `.qmd` file and your published `.html` report.

10.10 Reproducibility Practice

This assignment emphasizes transparent categorical analysis.

Your document must:

- Construct tables programmatically
- Avoid manual calculations outside your script
- Use clearly labeled code chunks
- Avoid hard-coding statistical results in narrative
- Use inline R expressions where appropriate
- Render cleanly to HTML

The goal is that another analyst could reproduce your entire Chi-Square analysis and verify every conclusion.

11 Wage Analytics — Predicting High vs Low Earners Using Logistic Regression

11.1 Overview

The **Wage Analytics Division (WAD)** has hired you as their new **Data Scientist**.

Your mission is to determine whether demographic and employment characteristics can be used to **predict whether a worker earns a high or low wage**.

Using the `Wage` dataset from the `ISLR` package, you will:

- Create a binary wage classification
- Explore relationships between predictors and wage
- Build a logistic regression model
- Evaluate predictive performance on unseen data

Your analysis will help the company understand which characteristics are associated with higher earnings and determine how well a predictive model performs in practice.

11.2 Learning Objectives

By completing this assignment, you will be able to:

- Clean and prepare mixed-type real-world data
- Create a binary outcome variable
- Conduct t-tests, ANOVA, and Chi-Square tests
- Perform a train/test split
- Build and interpret a logistic regression model
- Evaluate classifier performance using confusion matrices and ROC/AUC
- Communicate predictive modeling results clearly and professionally

---

11.3 Textbook Connection

This assignment builds directly from **Chapter 9: Logistic Regression** in Reproducible Research Using R.

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

11.4 Submission Instructions

Submit **two files**:

1. Your `.qmd` file
2. Your knitted and published `.html` file

Your document must:

- Knit without errors
- Suppress warnings and messages in the final HTML
- Include clearly labeled sections
- Include all statistical tests and model outputs
- Include performance metrics and ROC curve
- Include a professional final summary
- Use at least one inline R expression

11.5 Assignment Tasks

11.5.1 1. Create the WageCategory Variable

Begin by preparing the dataset.

11.5.1.1 Required Tasks

- Load the `Wage` dataset from the `ISLR` package
- Create a new factor variable called `WageCategory`:
  - If `wage > median(wage)` → `"High"`
  - If `wage < median(wage)` → `"Low"`
- Convert `WageCategory` into a factor variable

11.5.1.2 Important

- You must **NOT remove** `wage` or `logwage`
- Be prepared to explain the concept of **target leakage** and why it matters in predictive modeling

Briefly explain why creating a binary outcome transforms this into a classification problem.

11.5.2 2. Data Cleaning

Some categorical variables in the `Wage` dataset contain numeric prefixes (e.g., `"3. Asian"`).

11.5.2.1 Required Tasks

- Remove numeric prefixes from all relevant categorical variables
- Ensure categorical variables are clean and properly formatted before analysis

Provide a brief explanation of why cleaning categorical labels is important for interpretation and modeling.

11.5.3 3. Classical Statistical Tests

You will explore relationships between predictors and wage using classical inference methods.

11.5.3.1 A. t-test

Compare **age** between High and Low wage earners.

Your write-up must include:

- Mean age for each wage group
- t-statistic, degrees of freedom, and p-value
- A short interpretation of whether age differs between wage categories

---

11.5.3.2 B. ANOVA

Using the original numeric `wage` variable, choose **one** of the following:

- `wage ~ education`
- `wage ~ maritl`
- `wage ~ jobclass`

Your write-up must include:

- ANOVA table
- F-statistic, degrees of freedom, and p-value
- A brief interpretation of the result

11.5.3.3 C. Chi-Square Test

Choose **one categorical relationship** that is different from your ANOVA variable.

You must:

- Create a contingency table
- Conduct a Chi-Square test
- Report ^{2}, df, and p-value
- Compute and interpret Cramer's V
- Describe any patterns observed in the contingency table

Explain what the Chi-Square test tells you about association.

11.5.4 4. Logistic Regression Model

You will now build a predictive model.

11.5.4.1 Train/Test Split

- Perform a 70/30 split using `sample.split()`
- Clearly define your training and testing datasets

Explain why separating training and test data is critical in predictive modeling.

11.5.4.2 Model Specification

- Build a logistic regression model predicting `WageCategory`
- Choose predictors thoughtfully (be cautious of leakage or redundancy)

Your write-up must include:

- Logistic regression output
- Odds ratios using `exp(coef())`
- A 2–4 sentence interpretation describing:
  - Which predictors are significant
  - Direction of effects
  - Magnitude of effects
  - Any surprising findings

11.5.5 5. Model Evaluation on Test Data

Using your **test dataset**, compute:

- Predicted probabilities
- Predicted classes ("High" or "Low")
- Confusion matrix

Report and interpret:

- Accuracy
- Sensitivity
- Specificity
- Balanced Accuracy

11.5.5.1 ROC Curve and AUC

- Generate an ROC curve
- Compute AUC

In your interpretation, discuss:

- How well the model identifies high earners
- How accuracy compares to the No Information Rate
- Whether the model performs better than guessing

11.5.6 6. Final Interpretation

Write a professional 3–5 paragraph summary addressing:

- Whether age, education, jobclass, or marital status meaningfully relate to wage
- Which predictors were most influential in the logistic regression
- How well the model performed on unseen data
- Which wage group the model predicts best
- What variables you would add or remove if repeating the analysis
- One limitation of your modeling approach

Your tone should resemble a professional data science report, not a homework answer.

11.6 Guidelines for Interpretation

When interpreting regression results, include:

- Direction (positive or negative relationship)
- Magnitude (odds ratio interpretation)
- Statistical significance
- Plain-language meaning

Example:

> Workers with higher education levels have greater odds of being classified as high earners, holding other factors constant.

Avoid technical jargon without explanation.

11.7 Reflection

In 4–6 sentences, answer:

1. What is the difference between inference and prediction?
2. Why is target leakage dangerous in predictive modeling?
3. What did you learn about evaluating classifier performance?

11.8 Publishing Instructions

1. Render your `.qmd` file to HTML.
2. Confirm warnings and messages do not appear in the final document.
3. Submit both your `.qmd` file and your published `.html` report.

11.9 Reproducibility Practice

This assignment emphasizes responsible predictive modeling.

Your document must:

- Load data programmatically
- Clean variables reproducibly
- Use clearly labeled code chunks
- Avoid hard-coding values in narrative
- Use inline R expressions where appropriate
- Suppress warnings and messages in the final HTML
- Render cleanly without manual edits

The goal is that another analyst could reproduce your entire workflow and validate every statistical claim.

12 Final Project: NYC Open Data 2026 Proposal

12.1 Overview

Your final project for this course is to design and conduct an **original research project** using NYC Open Data.

Your project will be written as a fully reproducible **Quarto document** and structured as though it were ready for public presentation at **NYC Open Data Week**.

This assignment is designed to:

- Give you experience designing an independent research project
- Provide hands-on work with open civic datasets
- Produce a polished portfolio-ready artifact
- Strengthen your reproducible workflow and data communication skills
- Prepare you to submit to NYC Open Data Week 2026 if you choose

By the end of this course, you will have created a professional-quality research document suitable for public presentation.

12.2 Learning Objectives

By completing this project, you will be able to:

- Formulate a clear, data-driven research question
- Identify and justify appropriate open datasets
- Build a fully reproducible data workflow

- Clean, wrangle, and analyze civic data responsibly
- Create clear and meaningful visualizations
- Write for a public-facing, non-technical audience
- Connect your work to broader open data principles

12.3 Textbook Connection

This final project synthesizes concepts in Reproducible Research Using R.

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

12.4 Project Requirements

12.4.1 1. Dataset Selection

You must:

- Select **at least one dataset** from the NYC Open Data Portal (preferred).
- Other publicly available civic datasets regarding NYC are acceptable **with instructor approval**.
- Clearly cite and hyperlink all datasets used.
- Justify why the dataset is appropriate for your research question.

Your project should demonstrate thoughtful dataset selection rather than convenience-based selection.

12.4.2 2. Reproducible Analysis (Quarto)

Your final submission must:

- Be written in **Quarto (.qmd)** format.
- Include narrative, code, output, and interpretation in one document.
- Knit/render successfully without errors.
- Suppress unnecessary warnings and messages in the final output.

Your workflow should clearly demonstrate:

- Data import
- Cleaning and transformation
- Analysis aligned with your research question
- Transparent decision-making

12.4.3 3. Proposal-Style Structure

Your document should be structured as if it were an Open Data Week proposal.

Include the following sections:

12.4.3.1 A. Title & Event Description

- A clear, compelling project title.
- A 1–2 paragraph overview explaining:
  - Your research question
  - Why it matters
  - What attendees would learn

Write this as though it were submitted to Open Data Week.

12.4.3.2 B. Dataset(s) Used

- Describe the dataset(s).
- Provide proper citations and links.
- Explain how the dataset(s) connect to your research question.

12.4.3.3 C. Analysis

- Present a reproducible workflow from raw data to results.
- Clearly explain:
  - What you cleaned and why
  - What transformations were performed
  - What analyses were conducted
- Align every analysis step with your research question.

12.4.3.4 D. Visualizations

Include **at least two** clear, well-labeled figures or tables created in R.

All visualizations must:

- Include titles
- Include axis labels
- Include captions
- Be readable in a public-facing setting

Explain what each visualization shows and why it matters.

12.4.3.5 E. Audience & Relevance

Identify:

- Who would care about this project?
  - New Yorkers?
  - Policymakers?
  - Journalists?
  - Community organizations?
- Why does this research matter in a civic context?

12.4.3.6 F. Connection to Open Data

Explain how your project:

- Demonstrates transparency
- Highlights accessibility of public data
- Shows the value of open civic information
- Reflects open data principles

12.5 Length & Formatting Requirements

- Minimum **500 words of written narrative** (excluding code).
- Figures and tables must be clearly labeled.
- Citations must be properly formatted.
- The document must render cleanly to HTML or PDF.

Submit both:

- Your `.qmd` file
- Your knitted HTML or PDF output

12.6 Class Presentation

You will present your project during the final weeks of the semester.

Presentation Requirements:

- 10–12 minutes
- 2–3 minutes for Q&A
- Use slides (Quarto, PowerPoint, etc.)

Your presentation should:

- Clearly explain your research question
- Highlight key findings
- Show at least one visualization
- Be understandable to a non-technical audience

This is practice for a potential Open Data Week 2026 submission.

12.7 Class Publication (Optional but Encouraged)

At the end of the semester, projects may be compiled into a collective class book via Posit Cloud.

Each student's project may appear as a chapter.

Benefits:

- A publication credit
- A portfolio artifact
- A public-facing research contribution

You may opt out if you prefer.

12.8 Grading (100 Points Total)

- Dataset & Research Question (15 pts)
  Clarity, appropriateness, creativity
- Reproducibility (20 pts)
  Working document, organized workflow, transparent code
- Analysis (20 pts)
  Depth, correctness, alignment with question
- Visualizations (10 pts)
  Clarity, design, relevance
- Public Framing & Civic Relevance (10 pts)
  Event description, audience, connection to Open Data
- Class Presentation (15 pts)
  Clarity, engagement, professionalism
- Professionalism & Formatting (10 pts)
  Writing quality, citations, organization

12.9 Reproducibility Practice

This final project emphasizes transparent and repeatable workflows.

Your document must:

- Load all required packages explicitly
- Import data programmatically
- Avoid manual editing of datasets
- Clearly document cleaning decisions
- Avoid hard-coding results in interpretation
- Render cleanly without errors
- Set a seed if using sampling, modeling, or randomness

The goal is that another analyst could reproduce your findings exactly.

12.10 Resources

- NYC Open Data Portal: https://opendata.cityofnewyork.us/
- Past Open Data Week Events: https://2025.open-data.nyc/

12.11 Submission

Required:

- `.qmd` file
- Knitted HTML or PDF output

12.12 Takeaway

By completing this final project, you will leave the course with:

- An independent, reproducible research project
- A polished presentation
- A potential publication credit
- A strong foundation for an Open Data Week 2026 submission

13 Final Assignment: Building Your Reproducible Quarto Book

13.1 Overview

Your final assignment is to transform all of your completed course work into a polished, cohesive **Quarto book**.

Throughout the semester, every assignment was written in Quarto format for a reason. Now, you will combine them into a single published book.

You will:

- Convert your assignments into a structured Quarto book project
- Organize them into chapters
- Polish formatting and presentation
- Publish your completed book

This book will serve as a **professional portfolio artifact** that you can share with:

- Potential employers
- Graduate programs
- Research labs
- Collaborators
- Friends and family

By completing this assignment, you are not just submitting coursework —

You are officially becoming an author.

13.2 Learning Objectives

By completing this project, you will:

- Organize multiple Quarto documents into a unified book structure
- Apply consistent formatting and narrative across chapters
- Improve clarity, presentation, and documentation
- Practice professional-level data communication
- Publish a complete analytical portfolio

13.3 Textbook Connection

This final assignment synthesizes concepts in Reproducible Research Using R.

Students are encouraged to review the chapter before beginning this assignment, as it provides the conceptual foundation and reproducible workflow demonstrated here.

13.4 Assignment Requirements

13.4.1 1. Create a Quarto Book Project

You must:

- Create a Quarto book project (`_quarto.yml` configured as a book)
- Organize your assignments as individual chapters
- Ensure the book renders successfully from start to finish
- Publish your completed book (via your chosen publishing method)

Your book must render cleanly without errors.

13.4.2 2. Required Chapters (In This Order)

Your book must include the following chapters, in this exact order:

1. R Script → R Markdown Report
2. Homework: Continuing the Law Firm Analysis (Part 2)
3. Midterm
4. NBA Analytics – Exploring Team Performance Through Reproducible Analysis
5. Florida Crime Analytics – Uncovering the Root of Florida's Crime Surge
6. Streaming Analytics – Understanding Platform Popularity Across Age Groups
7. Wage Analytics – Predicting High vs Low Earners Using Logistic Regression
8. RMarkdown, Shiny, and NYC 311 Data

Each chapter must be its own `.qmd` file.

13.4.3 3. Title Page

Your book must include a professional title page containing:

- A clear, descriptive book title
- Your full name
- Course name and semester
- A short **"About This Book"** section (2–4 sentences)

This page should feel polished and intentional.

13.4.4 4. Cover Image

Your book must include a cover image.

Your cover image can be:

- Something you designed
- An image related to your analyses
- A professional photo or graphic
- A visual that represents your identity as a data analyst

The image must be appropriate and professional.

13.4.5 5. Chapter Introductions

Each chapter must begin with a short introduction explaining:

- The purpose of the assignment
- The skills or methods used
- Any relevant context for the reader

If the original assignment does not contain a chapter introduction, you must add one.

Think of this as guiding a new reader through your work.

13.4.6 6. Figure and Table Captions

Every figure, plot, and table must include:

- A clear, descriptive caption
- An explanation of what the reader is viewing
- Why the figure or table matters

Use proper Quarto chunk options (e.g., `fig-cap`) to ensure captions render correctly.

There should be no unlabeled figures in your book.

13.4.7 7. Final Reflection Chapter

Your final chapter must be a reflective essay (minimum 400 words) addressing:

- What you learned this semester
- What challenged you the most
- What you are most proud of
- How your skills and confidence have changed
- How reproducibility has influenced your thinking

This chapter should be written in thoughtful, professional prose.

13.5 Book Quality Standards

Your book should:

- Render cleanly from beginning to end
- Have consistent formatting across chapters
- Include clear headings and subheadings
- Avoid redundant code or messy output
- Suppress unnecessary warnings/messages
- Demonstrate attention to detail

This is not a collection of assignments.

It is a cohesive analytical portfolio.

13.6 Publishing Requirement

You must publish your completed Quarto book.

Your submission must include:

- A link to your published book
- Your complete Quarto book project folder

Your published book should be accessible and shareable.

13.7 Grading (100 Points Total)

- Structure & Organization (20 pts)
  Proper chapter order, functioning book structure
- Reproducibility (20 pts)
  Clean rendering, working code, no errors
- Presentation & Formatting (20 pts)
  Consistent style, captions, polished layout
- Chapter Introductions (10 pts)
  Clear context provided for each chapter
- Final Reflection (15 pts)
  Depth, thoughtfulness, clarity
- Professionalism (15 pts)
  Overall quality, cohesion, publication readiness

13.8 Reproducibility Practice

Your book must:

- Render entirely from source files
- Load all required packages explicitly
- Avoid hard-coded outputs
- Avoid manual edits to results
- Use consistent chunk labeling
- Be reproducible on another machine

The goal is that anyone could download your project and recreate your entire book.

13.9 Takeaway

By completing this assignment, you will leave the course with:

- A complete analytical book
- A published portfolio artifact
- Demonstrated mastery of reproducible workflows
- Authorship credit
- A powerful representation of your growth

This is the culmination of your semester's work.

Make it something you are proud to share.

A Undergraduate Midterm: Exercise & Sleep

A.1 Overview

You are a researcher in sleep expert **Dr. Matthew Walker's lab**. You have just run an experiment investigating how exercise relates to sleep habits.

You measured:

- Average hours of sleep **before** the experiment
- Average hours of sleep **after** the experiment
- **Sleep efficiency** at the end of the experiment

Dr. Walker wants you to investigate whether participants experienced changes in their sleep duration after participating in an exercise program.

He is also interested in whether these changes differ between males and females.

Your goal is to analyze the data and provide a **clear, evidence-based summary of your findings**.

This midterm assesses your ability to apply the skills we've learned so far and create a **reproducible research report using R and Quarto**.

A.2 Learning Objectives

By completing this midterm, you will be able to:

- Import data from a multi-sheet Excel workbook
- Clean and merge datasets
- Create new variables and identify missing values
- Calculate and interpret descriptive statistics
- Create clear visualizations using `ggplot2`
- Conduct and interpret t-tests
- Create a reproducible report using Quarto

- Incorporate inline R code and cross-references
- Cite R packages used in your analysis
- Communicate your findings through written interpretations

A.3 Textbook Connection

This assignment builds on the material covered in Reproducible Research Using R.

You are encouraged to use the textbook, previous assignments, and our in-class activities as references.

Remember: You do not need to memorize every function or command. Part of learning to code is learning how to find and apply examples!

A.4 Submission Instructions

Submit **three files**:

1. Your `.qmd` file
2. Your rendered `.html` file
3. Your `packages.bib` file

Your report must:

- Render without errors
- Include written explanations between code chunks
- Use clear section headings
- Include readable tables and visualizations
- Follow the Quarto requirements below

A.5 Quarto & Reproducibility Requirements

Your goal is to create a **professional research report**, not simply a collection of R code and statistical outputs.

A.5.1 1. YAML & Document Organization

Your report must include:

- A descriptive title
- Your name as the author
- A date
- A clickable table of contents
- Numbered sections

You are responsible for updating your YAML to include these features.

A.5.2 2. Code Chunk Labels

Every R code chunk must have a unique, descriptive label.

For example:

- `data-import`
- `data-cleaning`
- `descriptive-statistics`
- `ttest-sleep`

Additionally:

- Figure-producing chunks must begin with `fig-`
- Table-producing chunks must begin with `tbl-`

A.5.3 3. Figures, Tables & Cross-References

All figures and tables must include descriptive captions.

You must also **reference each figure and table in your written explanations** using Quarto cross-references.

For example:

- `@fig-sleep-change`
- `@tbl-sleep-summary`

Your written explanations should describe what the figures and tables show.

Remember!

When you use the appropriate labels and captions, Quarto automatically numbers your figures and tables.
Instead of manually writing "Figure 1", you can use `@fig-sleep-change` to create a reference that updates automatically.

A.5.4 4. Inline R Code

Your report must include **at least two instances of inline R code**.

Use inline R code to report values such as:

- The number of participants
- Average sleep duration
- Average sleep improvement
- Statistical results

Your written results should update automatically when your code is rerun.

A.5.5 5. Citing Your R Packages

An important part of reproducible research is acknowledging the software that makes our analyses possible.

For this midterm, you will practice **citing the R packages used in your analysis**.

You must:

1. Create a file called `packages.bib` containing references for the R packages you used.
2. Include your bibliography file in your Quarto YAML.
3. Cite the packages used in your analysis.
4. Make sure your references appear automatically when you render your report.

Need a Reminder?

You can use `knitr::write_bib()` to generate citations for your R packages.
For example:

```
knitr::write_bib(
  c("readxl", "dplyr", "ggplot2"),
  file = "packages.bib"
)
```

This is an example! Your bibliography should contain the packages you actually use. You will also need to add your bibliography to your Quarto YAML:

```
bibliography: packages.bib
```

Review the textbook section on citing R packages for additional guidance.

A.6 Midterm Tasks

A.6.1 1. Setup & Data Import

A.6.1.1 Tasks

- Load all packages you use.
- Import **both sheets** from `midterm_sleep_exercise.xlsx` using `readxl`:
  - `participant_info_midterm`
  - `sleep_data_midterm`
- Preview both datasets using an appropriate function, such as `glimpse()` or `head()`.

A.6.1.2 Requirements

- Use clearly labeled code chunks.
- Briefly describe the datasets you imported.

A.6.2 2. Merge & Base Cleaning

A.6.2.1 Tasks

- Clean column names so they are consistent and easy to work with.
- Standardize labels for:
  - `Exercise_Group` (e.g., “cardio”, “CARDIO” → “Cardio”)

  - `Sex` (e.g., “m”, “MALE” → “Male”)

- Merge the two datasets into one dataset.

A.6.2.2 Requirements

- Your final dataset should contain **one row per participant**.
- Your dataset should include participant information and sleep measurements.
- Show how many rows are in your merged dataset.
- Briefly explain how you merged the datasets.

A.6.3 3. Create a New Variable

A.6.3.1 Tasks

Create a variable called `Sleep_Difference` that represents the change in sleep duration:

```
Sleep_Difference = post_sleep - pre_sleep_num
```

A positive value indicates that a participant slept more after the experiment.

A negative value indicates that a participant slept less.

Next:

- Identify how many observations are missing `Sleep_Difference`.
- Remove observations with missing `Sleep_Difference`.

A.6.3.2 Requirements

- Show how many observations were removed.
- Briefly explain why you removed them.
- Use inline R code to report the number of observations removed.

A.6.4 4. Descriptive Statistics

A.6.4.1 Tasks

Calculate descriptive statistics for `Sleep_Difference`.

Include:

- Mean
- Standard deviation
- Minimum
- Maximum

Next, calculate the **mean Sleep_Difference by Sex**.

A.6.4.2 Requirements

- Present your results in clearly organized tables using `kable()` or an equivalent function.
- Include descriptive table captions using `tbl-cap`.
- Use the `tbl-` prefix for table-producing chunks.
- Reference your tables in your written explanation.

Write **2–3 sentences** describing what your results suggest.

A.6.5 5. Visualizations (2 Plots)

Create **two** visualizations:

1. **Boxplot:** `Sleep_Difference` by `Sex`
2. **Scatterplot:** `pre_sleep_num` vs. `post_sleep`, including a trend line

A.6.5.1 Requirements

Each visualization must include:

- A descriptive title
- Clear x- and y-axis labels
- A clean theme (e.g., `theme_minimal()`)
- A Quarto figure caption using `fig-cap`
- A unique chunk label beginning with `fig-`

After each visualization, write **1–3 sentences** describing what it suggests.

You must **reference each visualization in your written explanation** using Quarto cross-references.

For example:

"As shown in **?@fig-sleep-change**, …"

A.6.6 6. t-tests

Run the following **two t-tests**:

1. **Paired t-test:** Compare sleep duration before and after the experiment (`pre_sleep_num` and `post_sleep`).
2. **Independent-samples t-test:** Compare `Sleep_Difference` between males and females.

A.6.6.1 Requirements

For each t-test, report:

- Relevant group means
- t statistic
- Degrees of freedom
- p-value

Then write **2–3 sentences** interpreting each test.

Consider:

- Was the result statistically significant?
- What does the result suggest about sleep duration?
- Is the difference meaningful in practical terms?

> **i** Important!
>
> A statistically significant result does not automatically mean that a difference is large or practically important.
> Think about what your results mean in the context of sleep duration.

A.6.7 7. Synthesis & Recommendation

Write **4–6 sentences** answering:

Based on your analyses, what can you conclude about changes in sleep duration following the exercise program?

Your response should address:

- Did participants generally sleep more or less after the experiment?
- Was the change statistically significant?
- Did sleep improvements differ between males and females?
- What would you recommend Dr. Walker investigate next?

A.6.7.1 Requirements

- Support your conclusions using your descriptive statistics and t-tests.
- Reference at least one figure or table using Quarto cross-references.
- Include at least one instance of inline R code.
- Explain your findings in plain language.

Your recommendation should be understandable to someone who has not seen your R code.

> **Interpreting Your Findings**
>
> Remember that observing a change in sleep duration does not necessarily prove that exercise caused the change.
> Your conclusions should reflect what the data can actually tell us.

A.6.8 8. Reflection

Write **3–5 sentences** addressing:

- What was most challenging?
- What did you feel confident about?
- What would you do differently next time to improve your analysis or report?

A.7 Publishing Instructions

1. Render your `.qmd` file to HTML.
2. Confirm that your table of contents appears correctly.
3. Verify that your figures and tables are properly numbered and cross-referenced.
4. Confirm that inline R code displays the correct values.
5. Confirm that your package citations and references appear correctly.
6. Make sure your report renders without errors.
7. Submit your `.qmd`, `.html`, and `packages.bib` files.

A.8 Reproducibility Practice

This midterm evaluates both your statistical analysis and your ability to communicate research findings.

Your report must:

- Load all packages explicitly
- Import both sheets programmatically
- Merge the datasets correctly
- Handle missing values
- Use descriptive code chunk labels
- Include labeled figures and tables
- Reference figures and tables in your written explanations
- Use inline R code
- Cite the R packages used in your analysis
- Render successfully to HTML

Remember: Your goal is not simply to run statistical tests. Your goal is to create a reproducible research report that someone else can read, understand, and reproduce.

B Assignment 6: ANOVA and Payment Patterns in NYC Camera Violations

B.1 Overview

Congratulations! You've been hired as a data scientist for a law firm that specializes in fighting parking and camera tickets in New York City.

The firm wants to better understand patterns in NYC camera violations. Specifically, they want to know whether certain groups receive different fine amounts or make different payments.

Last week, we learned how to use **t-tests** to compare the means of two groups.

This week, we're taking things a step further!

You will use **Analysis of Variance (ANOVA)** to investigate differences across multiple groups using real-world NYC Open Data.

Your goal is to determine whether the differences you observe are statistically significant and whether they might be useful for the law firm.

Remember to include a hyperlink to the dataset through the NYC Open Data Portal in your introduction.

::: callout-note

B.2 Why Are We Learning ANOVA in Psychology?

Although today's dataset involves traffic violations, ANOVA is widely used in **psychology, sociology, and other social sciences**.

Researchers frequently want to compare outcomes across three or more groups.

For example:

- **Cognitive psychology:** Do students using three different study strategies earn different average exam scores?

- **Clinical psychology:** Do participants receiving different types of therapy report different average anxiety scores?
- **Social psychology:** Does average helping behavior differ across three experimental conditions?
- **Sociology:** Does average income differ across educational attainment groups?

In each example, researchers are comparing a **numeric outcome** across multiple groups.

ANOVA allows us to test whether there is evidence that at least one population group mean differs from the others.

The same statistical technique we use to compare NYC traffic violations can also help us answer questions about human behavior!
:::

B.3 Learning Objectives

By completing this assignment, you will be able to:

- Retrieve real-world NYC data using an R package.
- Clean and prepare data for statistical analysis.
- Identify independent and dependent variables.
- Conduct and interpret one-way ANOVAs.
- Create visualizations comparing multiple groups.
- Interpret F-statistics and p-values.
- Explore post-hoc comparisons using Tukey's HSD.
- Distinguish between statistical significance and practical importance.
- Present your findings in a reproducible Quarto document.

B.4 Textbook Connection

This assignment builds directly from **Chapter 5: Comparing Multiple Groups** in Reproducible Research Using R.

Review the chapter and our previous live coding scripts before beginning the assignment.

You may use these resources as references, but your analysis and written interpretations should be your own.

B.5 Submission Instructions

Submit the following:

1. Your completed `.qmd` file.
2. A link to your published HTML report on Posit Connect Cloud.
3. Your `packages.bib` bibliography file.

Your Quarto document must:

- Render successfully without errors.
- Include clearly labeled sections.
- Include at least one visualization.
- Include descriptive statistics and ANOVA results.
- Include written interpretations of your findings.
- Cite at least two R packages used in your analysis.
- Automatically generate a reference list.

Important: Unlike our in-class group activities, you are responsible for organizing and writing your own Quarto document.

B.6 Part 1: Load and Explore the Data

Begin by loading the NYC camera violations dataset using the `nycOpenData` package.

You may refer to our previous assignments and live coding scripts for examples.

B.6.1 Your Tasks

1. Load the necessary R packages.
2. Retrieve the NYC camera violations dataset.
3. Use `glimpse()`, `head()`, or `summary()` to explore the data.
4. Include a hyperlink to the dataset in your introduction.

B.6.2 Questions

Answer the following in your report:

1. What does this dataset represent?
2. How many observations are included?
3. Identify two variables that might be useful for investigating payment patterns.

::: callout-tip

B.7 Part 2: Prepare Our Data

Before conducting ANOVA, we need to make sure our variables are in the correct format.

For this assignment, we will focus on:

- `fine_amount`
- `payment_amount`
- `issue_date`
- `violation_type`

B.8 2A. Convert Numeric Variables

Use `mutate()` and `as.numeric()` to ensure that `fine_amount` and `payment_amount` are numeric.

Remove observations with missing values in the variables needed for each analysis.

Your Task: Include a `glimpse()` or `summary()` checkpoint showing that your numeric variables were converted successfully.

B.9 2B. Create Day of Week

Next, we want to investigate whether fine amounts differ depending on the day a violation was issued.

Using `issue_date`, create a new variable called `day_of_week`.

You may use the `lubridate` package.

::: callout-tip

B.10 Need a Hint?

Remember that `lubridate` includes useful functions for working with dates.

For example:

- `as.Date()` converts values into dates when the format is recognized.
- `wday()` can extract the day of the week.
- `filter()` can remove observations with missing values.

Make sure your dates are correctly formatted before extracting the day of the week.
:::

Your Task: Create a table showing how many observations belong to each day of the week.

B.10.1 Quick Check

Why is it important to make sure our variables are stored in the correct format before conducting statistical analyses?

Your Response:

B.11 Part 3: ANOVA Investigation 1 — Fine Amount by Day of Week

Let's begin our first ANOVA investigation!

The law firm wants to know:

> Do average fine amounts differ depending on the day of the week a camera violation was issued?

B.12 Step 1: Identify Your Variables

Before running ANOVA, identify:

- **Dependent variable:** _____
- **Independent variable:** _____

B.13 Step 2: Descriptive Statistics

Use `favstats()` or another method from class to calculate the average fine amount for each day of the week.

Questions:

1. Which day has the highest average fine amount?
2. Which day has the lowest average fine amount?

B.14 Step 3: Conduct a One-Way ANOVA

Use `aov()` to determine whether average fine amounts differ across days of the week.

::: callout-tip

B.15 ANOVA Reminder

The general structure is:

```
aov(numeric_variable ~ group_variable, data = dataset)
```

You can use `summary()` to examine the ANOVA results.
:::

B.15.1 Interpret Your Results

Answer the following:

1. What is the F-statistic?
2. What is the p-value?
3. Is the result statistically significant at the 0.05 level?
4. What does this suggest about fine amounts across days of the week?

::: callout-important

B.16 Statistical Significance vs. Practical Importance

A statistically significant result does not automatically mean that the difference is large or important.

With very large datasets, even small differences between group means can produce statistically significant results.

For example, imagine that average fines differ by only a few cents across days of the week.

That difference might be statistically significant, but would it really help a law firm make better decisions?

This is why we should consider both statistical significance and the actual size of the differences.
:::

B.17 Step 4: Explore Post-Hoc Comparisons

If your ANOVA is statistically significant, use `TukeyHSD()` to investigate which days differ from one another.

If your ANOVA is not statistically significant, explain why post-hoc comparisons may not be necessary for this investigation.

Question: Identify one statistically significant pairwise comparison, if any.

B.18 Step 5: Interpret Your Findings

Write a short paragraph (approximately 3–4 sentences) summarizing:

- Whether fine amounts differ significantly by day of week.
- Which days have higher or lower average fines.
- Whether the differences seem practically meaningful.

B.19 Part 4: ANOVA Investigation 2 — Payment Amount by Violation Type

Now let's investigate something that might be more useful to the law firm.

Instead of focusing on the amount of the fine, we will examine the amount actually paid.

Our research question is:

> Do average payment amounts differ across different types of camera violations?

This time, you will complete the analysis with less guidance.

B.20 Your Tasks

1. Identify the dependent and independent variables.
2. Calculate descriptive statistics for payment amounts by violation type.
3. Create a boxplot using `ggplot2`.
4. Conduct a one-way ANOVA using `aov()`.
5. Examine the F-statistic and p-value.
6. Interpret the results in a short paragraph.

Your boxplot must include:

- A descriptive title.
- Clearly labeled axes.
- Violation type on the x-axis.
- Payment amount on the y-axis.

B.20.1 Interpret Your Findings

Answer the following:

1. Which violation type has the highest average payment amount?
2. Is the ANOVA statistically significant?
3. Do the differences appear large enough to be practically meaningful?
4. Why might payment amounts be more useful to the law firm than fine amounts?

::: callout-note

B.21 Think Like a Researcher!

Remember that statistical significance tells us whether the data provide evidence of differences between group means.

It does not automatically tell us why those differences exist.

For example, differences in payment amounts could reflect differences in fine schedules, payment status, or other factors.

We should be careful not to interpret these findings as evidence that violation type directly causes different payment behavior.
:::

B.22 Part 5: Final Reflection

Now that you've completed both ANOVA investigations, imagine you are presenting your findings to the law firm.

Write a short reflection (approximately **4–6 sentences**) addressing:

1. What is the difference between statistical significance and practical importance?
2. Which of your two ANOVA investigations would be more useful to the law firm, and why?
3. What is one recommendation you would make based on your findings?
4. How could ANOVA be used to answer a research question in psychology or sociology?

For the final question, you may use one of the examples from the beginning of the assignment or develop your own.

B.23 Part 6: Citing Your R Packages

An important part of reproducible research is acknowledging the software that makes our analyses possible.

During our group activity, we learned how to generate a bibliography and cite R packages directly in Quarto.

Now it's your turn to apply those skills independently!

B.23.1 Your Tasks

1. Create a file called `packages.bib` containing references for the R packages used in your assignment.
2. Save it alongside your `.qmd` document.
3. Add `bibliography: packages.bib` to your Quarto YAML.
4. Include citations for at least **two R packages** used in your analysis.
5. Make sure your references appear automatically when you render your document.

::: callout-tip

B.23.2 Need a Reminder?

- `knitr::write_bib()` generates a bibliography.
- `citation()` displays citation information for R packages.
- `bibliography: packages.bib` connects your bibliography to Quarto.
- `[@R-ggplot2]` is an example of a citation key.

Remember to check the actual citation keys inside your `packages.bib` file!
:::

Your citations should appear naturally in your writing.

For example, you might cite the package used to retrieve the data or the package used to create your visualizations.

B.23.3 Example: Generating a Bibliography

We can use the `write_bib()` function from the `knitr` package to automatically generate citations for the R packages used in our analysis.

For example, suppose we used `ggplot2`, `dplyr`, and `nycOpenData` in our research.

We could run the following code in our **R Console**:

```
knitr::write_bib(
  c("ggplot2", "dplyr", "nycOpenData"),
  file = "packages.bib"
)
```

This code creates a file called `packages.bib` containing citation information for those three packages.

Notice that:

- `c()` contains the names of the packages we want to cite.
- `"packages.bib"` is the name of the bibliography file we are creating.
- We can add or remove package names depending on which packages we used in our analysis.

Once the file is created, we can connect it to our Quarto document by adding the following to our YAML:

```
bibliography: packages.bib
```

We can then cite our packages directly in our writing using citation keys, such as `[@R-ggplot2]`.

Your Turn: Modify the example above to generate a bibliography containing the R packages you used in this assignment.

> Remember!
>
> Always check the citation keys inside your `packages.bib` file before using them in Quarto. Your citation keys may differ depending on the package!

B.24 Final Checklist

Before submitting, make sure:

- My Quarto document renders successfully.
- I loaded and explored the NYC camera violations dataset.
- I cleaned the necessary variables.
- I completed both ANOVA investigations.
- I included descriptive statistics and a boxplot.
- I interpreted the ANOVA results.
- I completed the final reflection.
- I cited at least two R packages.
- My bibliography appears automatically.
- I submitted my `.qmd`, published report link, and `packages.bib` file.

Remember: Your goal is not simply to produce statistically significant results. Your goal is to conduct a reproducible analysis and explain what your findings actually mean.

C References

D References